From Source to Execution: Design and Implementation of a Statically-Typed Programming Language with Type Inference

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ABSTRACT

This report presents the design and implementation of a small, statically-typed interpreted programming language realised through a four-stage compiler pipeline. The language provides four primitive types — Integer, Float, Boolean, and String — and supports arithmetic and equality expressions, variable assignment with type inference from first use, conditional and iterative control flow, user-defined functions with value parameter passing, and a built-in `print()` function. Types are inferred from program context rather than declared explicitly, combining static type safety with a declaration-free syntax. The pipeline consists of a hand-written lexer, a recursive-descent parser that constructs an abstract syntax tree, a static type checker that performs inference and rejects ill-typed programs before execution, and a tree-walking interpreter. Both a command-line runner and an interactive desktop interface expose each pipeline stage for inspection. The report documents the formal grammar, the type inference algorithm, the system architecture, and the automated test suite of 59 tests that validates the complete implementation.

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CHAPTER 1

INTRODUCTION

Programming languages define the formal contract between a programmer's intent and machine execution. Building a language processor from source text through to evaluated output requires a coherent pipeline of components, each of which must correctly transform its input before passing it downstream. Lexical analysis identifies atomic tokens; parsing imposes hierarchical structure; static type checking validates semantic correctness before execution; and interpretation evaluates the validated program. When these stages are connected cleanly, errors are caught at the earliest possible point, and the overall system remains tractable to reason about, test, and extend.

This report presents the design and implementation of a small statically-typed interpreted language that realises this pipeline in full. The language supports four primitive types — Integer, Float, Boolean, and String — and provides arithmetic and equality expressions, assignment, conditional and iterative control flow, user-defined functions with value parameter passing, and a built-in `print()` statement. Types are inferred from program context rather than declared explicitly, combining the safety of static typing with the conciseness of a declaration-free syntax. The implementation spans three principal components developed in pipeline order: the lexer and symbol table, the recursive-descent parser and abstract syntax tree, and the type checker with interpreter. Each component was completed before the next was begun, since each stage consumes the structured output of its predecessor.

The implemented system can be invoked from the command line or through an interactive desktop interface, both of which surface the four pipeline stages — tokens, AST, inferred type table, and execution output — for inspection. An automated regression suite of 59 tests verifies correctness across all stages and language features.

The remainder of this report is organised as follows. Chapter 2 presents the language design, token specification, grammar, and operator precedence rules. Chapters 3 and 4 cover the front-end implementation in pipeline order: lexical analysis and the symbol table, then recursive-descent parsing and the AST. Chapter 5 explains the type inference algorithm and the semantic rules used by the checker. Chapter 6 presents the system architecture and implementation,

including the end-to-end pipeline, project layout, and technology stack. Chapter 7 discusses testing and verification, and Chapter 8 concludes the report.

CHAPTER 2

LANGUAGE DESIGN AND GRAMMAR

2.1 Types

The language provides four primitive types, chosen to cover the core domains of numeric computation, logical control flow, and text manipulation while keeping the type system tractable and the semantics well-defined.

2.1.1 Integer

The Integer type represents whole numbers such as `0`, `7`, and `42`. Integer literals are written as one or more decimal digits without quotation marks or a decimal point.

2.1.2 Float

The Float type represents real-valued numbers written with a decimal point, such as `3.14` or `0.5`. Float values are used exclusively with the dot-suffixed arithmetic operators (`+.`, `-.`, `*.`, `/.`), following the Caml Light convention. Integer division (`/`) performs floor division and returns `Integer`: for example, `10 / 3` evaluates to `3 : Integer`.

2.1.3 Boolean

The Boolean type represents logical truth values. The two Boolean literals are `true` and `false`. Boolean values are used in conditions and as results of comparison expressions.

2.1.4 String

The String type represents sequences of characters enclosed in double quotes, for example `"hello"`. Strings can be assigned to variables and printed through the built-in `print()` function. The surrounding quotes are stripped by the parser when the `Literal` node is created, so the value stored internally is the bare character sequence without surrounding quotes. At runtime, the built-in `print()` function re-adds double quotes around string values, so `print("plc")` outputs `"plc" : String`.

2.1.5 Void

Although the language exposes exactly four programmer-visible types, the implementation introduces a fifth value, `Void`, within the `LanguageType` enumeration. Its sole purpose is to provide a well-defined return type for functions that contain no `return` statement, allowing the symbol table to record a consistent entry for every function regardless of whether it produces a value. `Void` cannot appear in source code and is never visible to the programmer; it is a bookkeeping device used internally by the type checker.

2.2 Static Typing

The language employs static typing, meaning that type errors are detected at type-checking time rather than at runtime. No explicit variable type declarations are required. Instead, the type checker infers types from literals, assignments, operations, function calls, and return statements. Once a variable type is established, later assignments must remain consistent with that type. For example, the following program is rejected at the type-checking stage before any execution occurs:

```
x = 5;
x = "hello";   (* TypeError: Cannot assign String to variable 'x' of
   type Integer *)
```

This contrasts with dynamically-typed languages, where the same program would run without error until (or unless) a type-dependent operation was attempted.

2.3 Token Specification

Table 2.1 summarises the major token classes used by the language.

Table 2.1. Token Specification

Token Category	Example Lexemes
INT_LITERAL	0, 10, 42
FLOAT_LITERAL	3.14, 20.5
BOOL_LITERAL	true, false
STRING_LITERAL	"hello"
IDENTIFIER	variable and function names such as x, add
Arithmetic operators (integer)	+, -, *, /
Float arithmetic operators	+., -., *., /.
Comparison operators	==, !=
Assignment operator	=
Keywords	if, else, while, def, return, print
Delimiters	( ) { } , ;

2.4 Context-Free Grammar

The language grammar can be described by the context-free grammar $G = (V_N, V_T, P, S)$, where V_N is the set of non-terminals, V_T is the set of terminals, P is the set of production rules, and S is the start symbol `program`. The grammar is implemented through recursive-descent parsing functions.

2.4.1 Program Structure

```
program   -> statement* EOF
statement -> assignment
       | if_stmt
       | while_stmt
       | func_def
       | print_stmt
       | return_stmt
       | expr ';'
```

2.4.2 Arithmetic Expressions

```
expr   -> term (('+' | '-' | '+.' | '-.') term)*
term   -> factor (('*' | '/' | '*.' | '/.') factor)*
factor -> '-'  factor      (* integer unary negation *)
    | '-.' factor     (* float unary negation *)
    | INT_LITERAL
    | FLOAT_LITERAL
    | STRING_LITERAL
    | BOOL_LITERAL
    | IDENTIFIER
    | IDENTIFIER '(' args? ')'
    | '(' expr ')'
```

The operators `+`, `-`, `*`, and `/` operate exclusively on `Integer` operands. The dot-suffixed operators `+.`, `-.`, `*.`, and `/.` operate exclusively on `Float` operands. This follows the Caml Light convention of using distinct operator tokens for integer and float arithmetic, eliminating the need for implicit numeric promotion. Unary negation is similarly split: `-x` negates an `Integer` (desugared to `0 - x`), and `-.x` negates a `Float` (desugared to `0.0 -. x`).

2.4.3 Boolean Expressions

```
bool_expr -> expr '==' expr
       | expr '!=' expr
       | BOOL_LITERAL
       | expr          (* fallback: type checker enforces Boolean *)
```

Boolean expressions are only parsed as conditions inside `if` and `while` statements. Comparison results are not storable in variables directly; the grammar does not include `bool_expr` in the `factor` rule. This is consistent with the assignment specification, which describes Boolean expressions primarily as conditions.

If `bool_expr` parses an expression and finds no following `==` or `!=` operator, the expression is returned as-is and the type checker verifies that its inferred type is `Boolean`.

2.4.4 Statements

```
assignment  -> IDENTIFIER '=' expr ';'
if_stmt     -> 'if' '(' bool_expr ')' block ('else' block)?
while_stmt  -> 'while' '(' bool_expr ')' block
print_stmt  -> 'print' '(' expr ')' ';'
return_stmt -> 'return' expr ';'
block       -> '{' statement* '}'
```

2.4.5 Function Definitions and Calls

```
func_def -> 'def' IDENTIFIER '(' params? ')' block
params   -> IDENTIFIER (',' IDENTIFIER)*
args     -> expr (',' expr)*
```

Function calls use value parameter passing. This means the argument values are evaluated before the call and copied into the function's local environment.

2.5 Operator Precedence and Associativity

Operator precedence is encoded structurally in the recursive-descent parser. Comparison is lowest, addition and subtraction are next, and multiplication and division have the highest

precedence among binary operators. Arithmetic operators are left-associative.

Table 2.2. Operator Precedence (1 = lowest, 3 = highest) and Associativity

Precedence	Operators	Associativity
1 (lowest)	`==`, `!=`	non-associative (condition use only)
2	`+`, `-`, `+.`, `-.`	left-associative
3 (highest)	`*`, `/`, `*.`, `/.`	left-associative

CHAPTER 3

LEXICAL ANALYSIS AND SYMBOL TABLE

This chapter covers the responsibilities related to lexical and early semantic foundations of the language. It includes defining token structures and categories (such as keywords, operators, identifiers, and literals), implementing the lexer to transform source characters into tokens, and establishing the symbol table to manage variable and function storage along with their associated types. Together, these components ensure that raw source input is systematically organised into meaningful structures that can be used reliably by later stages of the compiler.

3.1 Token representation

The token system, written in `components/tokens.py`, establishes the vocabulary and structure of all lexical elements in the language. It specifies both the types of tokens that can exist and the structure each token must follow. This serves as a canonical contract between the lexer and the parser, ensuring consistency across all stages of the compiler.

Table 3.1 summarises the main categories, example lexemes, and typical `TokenType` values.

Table 3.1. Token categories and representative `TokenType` values

Category	Example lexemes	Representative `TokenType`
Literals	`42`, `3.14`, `true`, `"hello"`	`INT_LITERAL`, `FLOAT_LITERAL`, `BOOL_LITERAL`, `STRING_LITERAL`
Identifiers	`x`, `counter`, `add`	`IDENTIFIER`
Keywords	`if`, `else`, `while`, `def`, `return`, `print`	keyword-specific token variants
Integer operators	`+`, `-`, `*`, `/`	`PLUS`, `MINUS`, `STAR`, `SLASH`
Float operators	`+.`, `-.`, `*.`, `/.`	`PLUS_DOT`, `MINUS_DOT`, `STAR_DOT`, `SLASH_DOT`
Other operators	`=`, `==`, `!=`	`ASSIGN`, `EQUAL_EQUAL`, `BANG_EQUAL`
Delimiters	Parentheses, braces, comma, and semicolon (see lexer delimiter tokens)	delimiter token variants
End marker	end of source input	`EOF`

Token types (from class `TokenType`) are defined using an `Enum` with `auto()` to automatically assign unique values. This avoids reliance on raw strings and ensures safer comparisons dur-

ing parsing. The token categories include literals, identifiers, keywords, operators, delimiters, and a special end-of-file (`EOF`) token. The `EOF` token is appended after all input is processed, allowing the parser to detect the end of input explicitly rather than encountering an unexpected termination.

Each token is represented as an immutable dataclass (`frozen=True`), meaning its fields cannot be modified after creation. Each token contains four key pieces of information: its kind (`token_type`), its lexeme (the exact substring from the source code), and its source position (`line`, `column`). This structured representation allows later stages to process the program in terms of syntax and errors without handling raw text directly. Immutability improves reliability by preventing accidental modification during later processing.

```
@dataclass(frozen=True)
class Token:
   token_type: TokenType
   lexeme: str
   line: int
   column: int
```

3.2 Lexer implementation

The lexer, implemented in `components/lexica.py`, is a hand-written scanner that converts raw source text into a sequence of tokens. Unlike generated lexers (for example SLY lexer mode, or Lex and Flex), this implementation uses explicit control flow (`if` and `while`) to process input.

The lexer maintains internal state including the source string, current position, start of the current token, line number, and column index. The column index is tracked per line and resets upon encountering a newline, while a separate `token_column` records the starting position of each token for accurate error reporting.

The main function, `tokenize`, iterates through the source code, repeatedly invoking `_scan_token` to identify the next token. Tokens are appended to a list, and an `EOF` token is added at the end. Whitespace is skipped, while invalid input results in a `LexerError`.

```
def tokenize(self) -> list[Token]:
    tokens: list[Token] = []
    while not self._is_at_end():
        self.start = self.current
        self.token_column = self.column
        token = self._scan_token()
        if token is not None:
```

```
                tokens.append(token)
        tokens.append(Token(TokenType.EOF, "", self.line, self.column))
        return tokens
```

Token recognition follows a structured approach. Single-character tokens are mapped using a lookup table. Multi-character operators such as `==` and `!=` are handled using lookahead via `_match`. String literals are processed by `_string`, which reads characters until a closing quotation mark or the end of input is reached, with error handling for unterminated strings. Numeric literals are handled by `_number`, which distinguishes integers and floating-point values based on the presence of a decimal point followed by digits. Identifiers are processed greedily, consuming alphanumeric characters and underscores, and then checked against the `KEYWORDS` map to determine whether they are reserved words or user-defined names. Notably, `true` and `false` are also entries in this map, mapping to `TokenType.BOOL_LITERAL` rather than a keyword-specific type. This means boolean literals are recognised through the same keyword-fallback mechanism as reserved words, preventing them from being scanned as user-defined identifiers.

```
KEYWORDS: dict[str, TokenType] = {
    "if": TokenType.IF,
    "else": TokenType.ELSE,
    "while": TokenType.WHILE,
    # ...
    "true": TokenType.BOOL_LITERAL,
    "false": TokenType.BOOL_LITERAL,
}
```

```
# Arithmetic operators with optional trailing '.' for float variant
_ARITH_DOT: dict[str, tuple[TokenType, TokenType]] = {
    "+": (TokenType.PLUS,  TokenType.PLUS_DOT),
    "-": (TokenType.MINUS, TokenType.MINUS_DOT),
    "*": (TokenType.STAR,  TokenType.STAR_DOT),
    "/": (TokenType.SLASH, TokenType.SLASH_DOT),
}
```

When one of these characters is encountered, the lexer checks the next character: if it is a `.` (and the character after that is not a digit), the dot-suffixed float variant is emitted; otherwise the plain integer operator is emitted. This lookahead prevents the `.` in a float literal such as `3.14` from being consumed as part of an operator.

```
SINGLE_CHAR_TOKENS: dict[str, TokenType] = {
    "=": TokenType.ASSIGN,
    "(": TokenType.LPAREN,
    ")": TokenType.RPAREN,
    # ...
}
```

Helper functions such as `_advance`, `_peek`, and `_peek_next` manage character consumption and lookahead, while `_is_at_end` determines when input has been fully processed. Error messages include precise line and column information, aiding debugging and diagnostics.

Table 3.2 lists the main functions and data structures in `components/lexica.py` and their roles.

Table 3.2. Key functions and data in `lexica.py`

Function / data	Responsibility	Output / effect
`tokenize()`	Iterates through source, scans tokens, appends `EOF`	ordered token list
`_scan_token()`	Dispatches recognition based on character and lookahead	next token or skip
`_number()`	Parses numeric sequences, distinguishes int vs float	numeric literal token
`_string()`	Parses quoted text, checks termination	string token or error
`_identifier()`	Parses identifiers, resolves keyword vs user-defined	keyword or identifier token
`KEYWORDS` map	Maps reserved words to token types	token type selection
`SINGLE_CHAR_TOKENS` map	Maps single-character symbols	token type selection
`_advance()`, `_peek()`, `_match()`	Character consumption and lookahead	scanner state update

3.3 Symbol table and semantic structure

The symbol table, defined in `components/symbol_table.py`, operates at a higher level than the lexer. While the lexer processes raw text into tokens, the symbol table manages semantic information such as variable and function names, their types, and their scope.

The symbol table is implemented as a scoped structure, where each scope maintains its own dictionary of symbols and optionally references a parent scope. This enables proper handling of nested blocks and variable shadowing. Symbols are introduced through definition functions and retrieved through lookup operations, which search recursively through parent scopes if needed.

```
def lookup(self, name: str) -> Symbol:
    symbol = self._symbols.get(name)
    if symbol is not None:
        return symbol
    if self.parent is not None:
        return self.parent.lookup(name)
    raise SymbolTableError(f"Undefined symbol: {name}")
```

The type system is represented using a `LanguageType` enumeration, which includes fundamental types such as Integer, Float, Boolean, String, and Void. Symbols are modelled using an inheritance hierarchy: the base `Symbol` class stores a name and type, while subclasses such as `VariableSymbol` and `FunctionSymbol` include additional attributes. Variables track initialization state, while functions store parameter lists and return types. Table 3.3 summarises the main building blocks.

Table 3.3. Symbol table components and roles

Component	Stored information	Role in semantic analysis
Scoped symbol table	Symbols in current and parent scope	Supports nesting and shadowing via recursive lookup
`LanguageType` enum	Integer, Float, Boolean, String, Void	Defines the type system
`Symbol` (base)	Name and declared or inferred type	Base abstraction for entries
`VariableSymbol`	Variable metadata (including initialization state)	Tracks correct usage before assignment
`FunctionSymbol`	Parameters and return type	Validates calls and return consistency
`lookup(name)`	Recursive scope search	Resolves identifiers or reports undefined
`SymbolTableError`	Semantic error details	Reports duplicates, undefined symbols, invalid operations

Errors related to semantic violations are handled via `SymbolTableError`. These include duplicate declarations within the same scope, references to undefined symbols, and invalid operations such as attempting to mark a non-variable symbol as initialized.

3.4 Role of the symbol table in the pipeline

The symbol table is constructed during the type-checking phase rather than during lexical analysis. This guideline is followed in the implementation. The type checker initialises a global symbol table and populates it while traversing the AST, including defining variables and functions, managing nested scopes, and validating type correctness.

```
def run_pipeline(source_code: str) -> PipelineResult:
    tokens = Lexer(source_code).tokenize()
    tree = Parser(tokens).parse()
    ast_text = ASTPrinter().print(tree)
    checked_scope = TypeChecker().check(tree)
    outputs: list[str] = []
    Interpreter(output_callback=outputs.append).execute(tree)
    return PipelineResult(
        tokens=tokens,
```

```
        ast_text=ast_text,
        checked_scope=checked_scope,
        outputs=outputs,
    )
```

Once type checking is complete, the populated symbol table is returned as part of the pipeline result (`checked_scope`). The checker creates child scopes for blocks and functions during analysis, but the displayed `checked_scope` table represents the global scope entries returned by the pipeline. It can then be displayed or used for further analysis.

This separation of concerns keeps lexical analysis focused on tokenization, while semantic meaning is enforced during type checking. For user-defined functions, definitions are first registered in the global scope with placeholder parameter and return types. These are inferred from the first valid call and refined through body checking; subsequent calls must match the inferred parameter types.

CHAPTER 4

RECURSIVE-DESCENT PARSING AND THE ABSTRACT SYNTAX TREE

This chapter presents the implementation of the recursive-descent parser and the associated abstract syntax tree (AST) model. Starting from lexer-produced tokens, the parser constructs a hierarchical `Program` AST that represents expressions, statements, and control-flow structure in a form suitable for downstream semantic analysis and execution. Implemented syntax coverage includes arithmetic expressions, assignments, if-then-else, while, function definitions, function calls, return, and built-in `print(...)` statements. Equality operators (`==`, `!=`) are parsed in condition contexts (if/while) through the parser's boolean-expression path.

4.1 Abstract syntax tree representation

The AST is defined as a collection of Python dataclasses representing the structural elements of the programming language. It serves as an intermediate representation between parsing and later stages such as type checking and interpretation.

A key distinction exists between tokens and AST nodes. Tokens are flat and describe individual lexical elements such as operators or identifiers. In contrast, the AST is hierarchical. For example, while a token may represent the symbol `+`, a `BinaryOp` node represents an entire addition operation, including its left and right operands as subtrees.

All AST nodes inherit from a common base class, allowing them to store positional metadata (`line`, `column`). This ensures that errors detected in later stages can still reference precise locations in the original source code.

```
@dataclass
class ASTNode:
    """Base class for every AST node. Carries source location for
       diagnostics."""
    line: int = field(default=0, kw_only=True)
    column: int = field(default=0, kw_only=True)
```

4.1.1 *Expressions*

Expressions represent computations or values within the program. A `Literal` node stores values known at parse time, including integers, floating-point numbers, booleans, and strings. String literals are stored without surrounding quotation marks, as these are removed during parsing.

```
if tok.token_type == TokenType.STRING_LITERAL:
    self._advance()
    return Literal(value=tok.lexeme[1:-1], line=tok.line, column=tok.
        column)
```

An `Identifier` node represents the use of a variable name within an expression, indicating evaluation rather than declaration. A `BinaryOp` node models arithmetic and comparison operations such as addition, subtraction, multiplication, division, and equality checks, storing the operator along with references to left and right operands. A `FunctionCall` node represents function invocation, consisting of the function name and a list of argument expressions.

```
@dataclass
class BinaryOp(ASTNode):
    """An arithmetic or comparison binary operation.

    op is one of: '+' '-' '*' '/' '+.' '-.' '*.' '/.' '==' '!='
    Example:  a + b    x == 0    a +. b
    """
    op: str
    left: ASTNode
    right: ASTNode
```

4.1.2 *Statements*

Statements represent actions or control flow within the program. An `Assign` node models variable assignment, pairing a variable name with a value expression, while `Print` and `Return` nodes each store a single expression and represent output and function return operations respectively. A `Block` node groups multiple statements within braces, representing a scoped execution sequence.

Control flow is handled via `If` and `While` nodes. The `If` node stores a condition, a then-block, and an optional else-block, while `While` stores a loop condition and body block. Function definitions are represented by `FunctionDef` nodes, which include the function name, a list of parameter names, and a body block. Parameter types are intentionally not assigned at parse time, as type inference is handled later by the type checker. Finally, `Program` serves as the

AST root, containing all top-level statements.

4.2 Parser implementation

The parser consumes the list of tokens generated by the lexer and produces a single `Program` AST node. It is implemented as a hand-written recursive-descent parser based on the project grammar. This avoids parser generators and instead relies on explicit functions for each grammar rule.

The parser maintains internal state consisting of the token list and a position index (`self._pos`) indicating the current token. Tokens are accessed via helper methods such as `_peek` and `_advance`. Parsing errors are reported using a custom `ParseError` exception with line and column information consistent with lexer diagnostics.

```
def parse(self) -> Program:
    """Parse the full token stream and return the root Program node."""
    line, col = self._peek().line, self._peek().column
    statements: list[ASTNode] = []
    while not self._is_at_end():
        statements.append(self._statement())
    return Program(statements=statements, line=line, column=col)
```

In addition to the main `parse()` entry point, the parser's statement dispatcher (`_statement`) implements a deterministic decision order over token patterns: function definition, conditional, loop, return, print, assignment, then expression statement fallback. This design is important because it prevents ambiguity between syntactically similar forms. In particular, assignments are recognized using one-token lookahead (`IDENTIFIER` followed by `=`), while identifier-led expressions that are not assignments are parsed as expression statements and must still end with a semicolon. This gives the parser full coverage of both declarative-style and expression-driven top-level statements without requiring a parser generator.

4.2.1 Statement parsing

The main parsing loop repeatedly processes statements until `EOF`, with the statement type determined via dispatch on the current token. Function definitions, conditionals, loops, returns, and prints each have dedicated parsing methods. Assignments are detected using one-token lookahead (`IDENTIFIER` followed by `ASSIGN`), preventing misclassification of function calls as assignments. If no specific statement pattern matches, the parser falls back to an expression statement and enforces a terminating semicolon.

```
# assignment: IDENTIFIER "=" expr ";"
if (
    tok.token_type == TokenType.IDENTIFIER
    and self._peek_next().token_type == TokenType.ASSIGN
):
    return self._assignment()
# fallback: bare expression statement  expr ";"
node = self._expr()
self._expect(TokenType.SEMICOLON, "Expected ';' after expression")
```

4.2.2 Expression parsing and precedence

Expressions are parsed in layered precedence levels: `_expr` handles `+`, `-`, `+.`, and `-.`; `_term` handles `*`, `/`, `*.`, and `/.`; and `_factor` handles literals, identifiers and function calls, and parenthesized expressions. Boolean conditions for `if` and `while` are parsed by `_bool_expr`, which recognises `==` and `!=` after parsing the left-hand arithmetic expression. As a result, equality parsing is condition-focused in the current grammar implementation rather than a general infix operator available in every expression position. Function calls are recognised when an identifier is followed by `(`, with argument lists parsed as comma-separated expressions. Parenthesized expressions are supported to override precedence. Unary negation (`-x`, `-5`) is also handled at the `_factor` level: a leading `-` token causes the parser to recursively parse the operand and return `BinaryOp(0 - operand)`, so the existing type rules for subtraction cover negation without any additional inference code.

```
def _expr(self) -> ASTNode:  # +,-
    ...
def _term(self) -> ASTNode:  # *,/
    ...
def _factor(self) -> ASTNode:  # unary -, literals, identifiers/calls,
    grouped expr
    ...
```

Figure 4.1 shows the AST produced for the statement `z = x + y * 2;`. Because `_term` handles `*` before `_expr` handles `+`, the multiplication sub-tree is nested one level deeper, encoding the higher precedence of `*` over `+` without any explicit priority table at runtime.

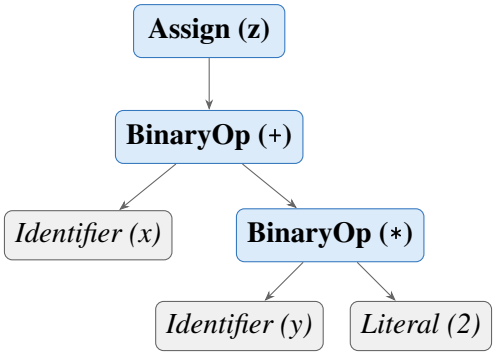


Figure 4.1. AST for `z = x + y * 2;`. Multiplication is a deeper sub-tree than addition, encoding higher precedence through tree structure rather than a runtime priority table.

Expression parsing is intentionally split into layered methods to enforce precedence and associativity in a transparent way. Parenthesized expressions are parsed explicitly and re-enter `_expr`, allowing precedence override where required by source syntax. Boolean conditions for `if` and `while` are parsed through `_bool_expr`, which accepts equality and inequality comparisons but can also return a non-comparison expression node; semantic enforcement that a condition is truly Boolean is deferred to the type checker.

4.2.3 Helper functions and lookahead

As described above, the parser uses `_check`, `_expect`, `_peek`, `_peek_next`, `_advance`, and `_is_at_end` for token navigation and validation. Lookahead is central for ambiguity resolution: assignment versus function call is resolved by inspecting the next token; comparison operators are recognised after parsing the left expression in `_bool_expr`. Parser errors follow a consistent diagnostic format: `[line X, col Y] ParseError: <message>`.

The parser also includes dedicated list parsers for formal parameters (`_params`) and call arguments (`_args`), both implemented as comma-separated sequences with optional emptiness. This ensures consistent handling of signatures such as `def f()` and `def f(a, b)` as well as invocations like `f()` and `f(x, y+1)`.

4.3 AST printing

For debugging and reporting, an AST printer converts the tree into a readable indented string, integrated into the pipeline output. The printer uses visitor-style dynamic dispatch, mapping node class names to methods such as `_visit_If` and `_visit_BinaryOp`, with output built as multi-line text where indentation reflects tree depth. Node formatting emphasises structure:

`Program` and `Block` include statement counts, `BinaryOp` explicitly shows left and right subtrees, and `FunctionCall` prints each argument by index. `Literal` includes both value and runtime Python type names for clarity.

4.4 Parser and type checker boundary

A deliberate architectural boundary is maintained between syntax construction and semantic validation. The parser is responsible for producing a structurally valid AST with accurate source locations, while type correctness and semantic constraints are checked later by the type checker. For example, `_bool_expr` may syntactically accept a non-Boolean expression as a condition node, but rejection of non-Boolean control-flow conditions is handled during semantic analysis. This separation keeps parser logic focused on grammar conformance and allows a clear responsibility split across team member contributions.

CHAPTER 5

TYPE INFERENCE ALGORITHM

5.1 Overview

Type inference in this project means that the programmer does not write explicit type declarations for variables or functions. Instead, the compiler infers types from assignments, literals, operations, function calls, and return statements. This keeps the language simple while still preserving static type safety.

The type checker runs after parsing. It receives the AST and validates the program before execution. Any inconsistent use of types is reported as a type error, and execution is stopped.

The interpreter stage does not repeat this validation: it walks the same AST with ordinary Python representations of values. Static type information remains in the symbol table produced by the checker; run-time values are not separately labelled with `LanguageType`, except that `print()` formats output by inspecting the value's Python type.

5.2 Type Environment

The type environment is represented by the symbol table. Conceptually, it is a mapping

$$\Gamma : \textit{Identifier} \rightarrow \textit{Type}$$

The environment grows as the checker visits the AST. On first assignment, a variable name is inserted with its inferred type. When a new block or a function call is entered, a child environment is created so that nested scopes can be checked independently while still accessing outer bindings when needed.

5.3 Type Inference Procedure

The type checker performs type inference by traversing the abstract syntax tree before interpretation. The algorithm is syntax-directed: each AST node is visited, the types of its child nodes are inferred first, and the current node is checked according to the language typing rules.

Algorithm TYPECHECK($node$, Γ)

Input: *node* = current AST node; Γ = current type environment / symbol table

Output: inferred type of *node*, or `TypeError`

1. **If** *node* is a **Literal**: return the corresponding primitive type: `Integer`, `Float`, `Boolean`, or `String`.
2. **If** *node* is an **Identifier**: look up the identifier in Γ. If it is not found, report an undefined variable error. Otherwise, return its stored type.
3. **If** *node* is a **BinaryOp**:
   - Infer the type of the left operand and the right operand.
   - If the operator is `+`, `-`, `*`, or `/`: require both operands to be `Integer`; return `Integer`.
   - If the operator is `+.`, `-.`, `*.`, or `/.`: require both operands to be `Float`; return `Float`.
   - If the operator is `==` or `!=`: require both operands to be the same arithmetic type; return `Boolean`.
   - Otherwise, report a `TypeError`.
4. **If** *node* is an **Assignment**:
   - Infer the type of the right-hand-side expression.
   - If the variable is not yet in Γ: insert it with the inferred type.
   - Else: verify the new type matches the existing type; if not, report a `TypeError`.
5. **If** *node* is an **If** statement:
   - Infer the condition type; require it to be `Boolean`.
   - Type-check the then-block in a child environment.
   - If an else-block exists, type-check it in a child environment.
6. **If** *node* is a **While** statement:
   - Infer the condition type; require it to be `Boolean`.
   - Type-check the loop body in a child environment.
7. **If** *node* is a **FunctionDef**:
   - Register the function name in Γ.
   - Parameter types are initially unknown.
   - Return type is inferred from `return` statements in the body.

8. **If** *node* is a **FunctionCall**:
   - Infer all argument types.
   - If this is the first valid call: store the argument types as the function's parameter types.
   - Else: compare argument types with the stored parameter types.
   - Type-check the function body using a local environment.
   - Return the inferred function return type.
9. **If** *node* is a **Print** statement: infer the expression type; accept any of the four supported types.

5.4 Inference Rules

5.4.1 Literals

Literal nodes have direct types:

$$\Gamma \vdash 42 : \texttt{Integer}$$

$$\Gamma \vdash 3.14 : \texttt{Float}$$

$$\Gamma \vdash \texttt{true} : \texttt{Boolean}$$

$$\Gamma \vdash \text{“}\texttt{hi}\text{”} : \texttt{String}$$

5.4.2 No Operator Overloading

The language does not support operator overloading. Following the Caml Light convention, integer and float arithmetic use entirely separate operator tokens, so every operator has a single, fixed type signature:

- `+`, `-`, `*`, `/` — require two `Integer` operands; result is `Integer`.
- `+.`, `-.`, `*.`, `/.` — require two `Float` operands; result is `Float`.
- `+` is not overloaded for string concatenation; applying it to a `String` is a type error.
- `==` and `!=` are not overloaded for `Boolean` or `String` equality.
- No implicit numeric promotion: `Integer + Float` is a type error.

Because every operator has a unique, unambiguous type derivable from its token alone, the compiler can always infer expression types without requiring explicit type declarations from the programmer.

5.4.3 Arithmetic Expressions

The integer operators `+`, `-`, `*`, and `/` require both operands to be `Integer` and always return `Integer` (including division, which uses integer floor division). The float operators `+.`, `-.`, `*.`, and `/.` require both operands to be `Float` and always return `Float`. Mixing an `Integer` and a `Float` with any operator is a type error.

Table 5.1. Arithmetic operator type rules

Left operand	Operator	Right operand	Result type
`Integer`	`+`, `-`, `*`, `/`	`Integer`	`Integer`
`Float`	`+.`, `-.`, `*.`, `/.`	`Float`	`Float`
`Integer`	any	`Float`	`TypeError`
`Float`	any	`Integer`	`TypeError`
Any	any	`String` or `Boolean`	`TypeError`

Unary negation

Unary integer negation (`-x`) is desugared by the parser to `0 - x`. Unary float negation (`-.x`) is desugared to `0.0 -. x`. The type checker therefore applies the standard binary rules in both cases.

5.4.4 Boolean Expressions

The comparison operators `==` and `!=` always produce `Boolean`. Both operators require two operands of the *same* arithmetic type — either both `Integer` or both `Float`; mixed-type comparisons such as `Integer == Float` are rejected. Comparisons involving `Boolean` or `String` operands are also rejected by the type checker. Although `Boolean` and `String` values exist in the language, equality and inequality are deliberately restricted to arithmetic expressions because the project specification defines basic Boolean expressions as equality and inequality between two arithmetic expressions.

5.4.5 Assignment Statement

Assignments define variable types on first use. If a variable does not yet exist in the type environment, the checker infers the type of the right-hand side and inserts the variable into the

environment. If the variable already exists, the new value must have the same type.

$$\text{If } x \notin \Gamma \text{ and } \Gamma \vdash e : T, \text{ then } \Gamma, x : T \vdash x = e$$

$$\text{If } x : T \in \Gamma \text{ and } \Gamma \vdash e : T, \text{ then } \Gamma \vdash x = e$$

5.4.6 If-Then-Else

The condition of an `if` statement must be `Boolean`. The then-block and else-block are checked in child scopes so that local operations can be validated cleanly.

$$\Gamma \vdash c : \texttt{Boolean} \quad \Gamma' \vdash B_{then} \quad \Gamma'' \vdash B_{else} \quad \Longrightarrow \quad \Gamma \vdash \texttt{if}\,(c)\,B_{then}\,\texttt{else}\,B_{else}$$

5.4.7 While-Loop

The condition of a `while` loop must also be `Boolean`. The loop body is checked in a child scope and must not violate the types of variables already introduced in the surrounding environment.

$$\Gamma \vdash c : \texttt{Boolean} \quad \Gamma' \vdash B \quad \Longrightarrow \quad \Gamma \vdash \texttt{while}\,(c)\,B$$

5.4.8 Function Definition and Call

Function parameter types are inferred from the first call to the function. After that first call, subsequent calls must use arguments of the same types. The return type is inferred from the `return` statements in the function body. If the function returns inconsistent types, the checker reports an error.

For functions that are defined but never called, the type checker performs a post-pass after all top-level statements have been processed. It infers parameter types by scanning the body for arithmetic and comparison usage, then checks the full body with those inferred types. This ensures that type errors inside never-called function bodies are still detected before execution.

Recursive calls are intentionally outside the scope of this implementation; supporting them would require additional fixed-point inference or explicit recursion handling that is orthogonal to the core type-inference algorithm.

5.5 Type Error Reporting

Type errors are reported with source positions so that the user can locate the problem quickly. The format used by the system is:

```
[line X, col Y] TypeError: message
```

Examples include assigning a `String` to an `Integer` variable, using a non-Boolean condition in an `if` statement, or supplying the wrong argument type to a function.

5.6 Example Type Inference Traces

Consider the following program fragment:

```
x = 3;
y = 4;
z = x + y;
print(z);
```

The checker infers:

- `x`: `Integer` (from literal 3)
- `y`: `Integer` (from literal 4)
- `x + y`: `Integer` (Integer + Integer → Integer)
- `z`: `Integer`

For a float example using dot-suffixed operators:

```
def addf(a, b) {
        return a +. b;
}

result = addf(2.0, 3.5);
```

The first function call infers `a`: `Float` and `b`: `Float`. The operator `+.` requires both operands to be `Float`, so the function return type is inferred as `Float`. The variable `result` is also inferred as `Float`.

CHAPTER 6

SYSTEM ARCHITECTURE AND IMPLEMENTATION

This chapter describes how the implementation is organised: the end-to-end processing flow, repository layout, technology choices, and the main components that realise static analysis and execution (type checker, interpreter, and user interfaces). Detailed treatment of lexical analysis, the symbol table, and parsing appears in Chapters 3 and 4; Chapter 5 presents the type-inference rules that the type checker implements.

6.1 Compiler pipeline

The system follows a four-stage pipeline. Lexical and syntactic processing are kept separate from semantic analysis and execution, which simplifies testing and localises errors to the appropriate stage.

The overall flow of data through the system is shown in Figure 6.1.

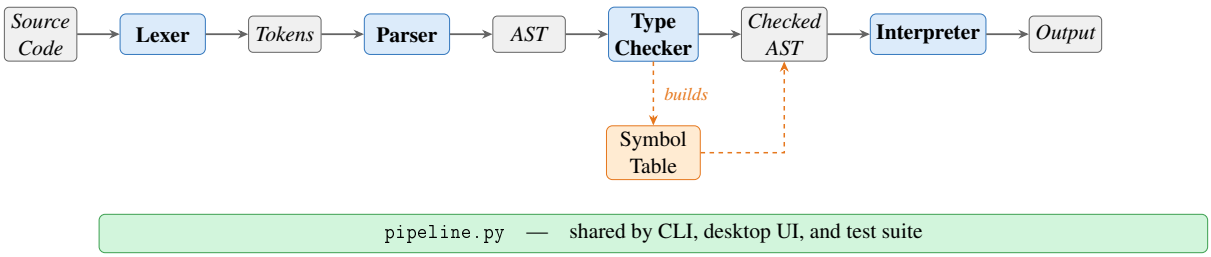


Figure 6.1. Compiler pipeline overview.

Source characters are scanned by the lexer into a token stream. The parser consumes tokens and builds an AST that reflects program structure (expressions, statements, and control flow). The type checker then walks the AST, populates and uses the symbol table, infers types, and validates scope and type rules. Only programs that pass the type checker proceed to interpretation. The interpreter evaluates the checked tree using ordinary Python values; it does not re-run the type rules, since correctness at runtime is guaranteed by the prior static pass. All four stages are orchestrated through `pipeline.py`, which is shared by the command-line runner, the desktop UI, and the test suite.

6.1.1 End-to-end pipeline walkthrough

The following example traces the complete execution of a three-statement program through all four pipeline stages, illustrating the correspondence between source text, tokens, abstract syntax tree, inferred types, and runtime output.

Source program:

```
a = 3;
b = 4;
print(a + b);
```

Stage 1 — Lexer output (token stream):

```
IDENTIFIER     'a'      line 1, col 1
ASSIGN         '='      line 1, col 3
INT_LITERAL    '3'      line 1, col 5
SEMICOLON      ';'      line 1, col 6
IDENTIFIER     'b'      line 2, col 1
ASSIGN         '='      line 2, col 3
INT_LITERAL    '4'      line 2, col 5
SEMICOLON      ';'      line 2, col 6
PRINT          'print'  line 3, col 1
LPAREN         '('      line 3, col 6
IDENTIFIER     'a'      line 3, col 7
PLUS           '+'      line 3, col 9
IDENTIFIER     'b'      line 3, col 11
RPAREN         ')'      line 3, col 12
SEMICOLON      ';'      line 3, col 13
EOF                     line 3, col 14
```

Stage 2 — Parser output (abstract syntax tree):

```
Program  [3 statement(s)]
    Assign  'a'  =
        Literal  3  (int)
    Assign  'b'  =
        Literal  4  (int)
    Print
        BinaryOp  '+'
            left:
                Identifier  'a'
            right:
                Identifier  'b'
```

Stage 3 — Type checker output (inferred type environment):

Name	Inferred type	
`a`	`Integer`	
`b`	`Integer`	
`a + b`	`Integer`	(Integer + Integer → Integer)

Stage 4 — Interpreter output:

```
7 : Integer
```

The example demonstrates the full data flow: source characters are tokenised by the lexer, structured into an AST by the parser, validated and annotated by the type checker, and finally evaluated by the interpreter to produce typed output.

6.1.2 Entry points and shared pipeline

All stages are orchestrated by `components/pipeline.py`, which is shared by the command-line runner, the desktop UI, and the test suite. From the project root, `python3 main.py` runs a script (or built-in sample) and prints the stages in order: tokens, AST text, type table, and execution output. `python3 ui.py` runs the Tkinter workbench: the same pipeline is used internally, with each stage shown in a separate tab so the user can inspect results without altering the underlying flow.

6.2 Project structure

The main source files and their responsibilities are summarised in Table 6.1.

Table 6.1. Main source files

File	Responsibility
`main.py`	Command-line runner for sample code or a script file
`ui.py`	Desktop UI for editing and running a script
`components/tokens.py`	Token classes and token categories
`components/lexica.py`	Lexical analysis
`components/ast_nodes.py`	AST node definitions
`components/parser.py`	Recursive-descent parser
`components/ast_printer.py`	AST renderer for debugging and reporting
`components/symbol_table.py`	Scoped symbol table and semantic types
`components/type_checker.py`	Static type checker and inference
`components/interpreter.py`	Tree-walking interpreter
`components/pipeline.py`	Shared pipeline for CLI, UI, and tests

6.3 Technology stack

The project uses Python and standard-library tools only (Table 6.2).

Table 6.2. Technology stack

Tool	Purpose
Python 3.10+	Core implementation language
`dataclasses`	AST and runtime helper structures
`tkinter`	Desktop graphical interface
`unittest`	Automated testing

6.4 Type checker

The `TypeChecker` class implements the inference and checking rules described in Chapter 5. It traverses the AST before execution, uses the symbol table to record variable and function information, infers types on first assignment, validates arithmetic and comparison expressions, enforces Boolean conditions for control flow, infers function parameter types from the first call (or from body usage for uncalled functions), and infers function return types from `return` statements. Programs that fail these rules are rejected before interpretation.

6.4.1 Scope of the implemented type system

In scope terms, this component turns a syntactically valid AST into a semantically valid program:

- static typing rules for arithmetic, comparison, assignment, control flow, and functions,
- type checking for expressions and statements,
- variable type inference from first assignment,
- function parameter inference from the first call (or from body usage for uncalled functions),
- function return type inference from `return` statements,
- source-positioned type errors when a program violates a typing rule.

6.5 Interpreter

The interpreter is a tree-walking evaluator over the checked AST. It maintains nested runtime environments for variables and function calls. Assignments write through the current or nearest enclosing scope; `if` selects a branch; `while` repeats until its condition becomes false; function calls bind arguments by value and return through an internal return mechanism.

The built-in `print()` function outputs both value and runtime type. All four types are supported:

```
print(42);        (* 42 : Integer        *)
print(3.14);      (* 3.14 : Float        *)
print(true);      (* true : Boolean      *)
print("plc");    (* "plc" : String      *)
```

6.5.1 Scope of the implemented runtime

The runtime stage provides observable behaviour and integrates with the shared pipeline:

- assignment execution,
- `if` execution,
- `while` execution,
- function execution with value parameter passing,
- `print()` execution with immediate output,
- execution behind the same `run_pipeline` path used by the CLI, UI, and tests.

6.6 Graphical user interface

The desktop workbench (`ui.py`) is implemented with Python's `tkinter` standard library and requires no additional dependencies. The window is titled *125970-126690-126687-Ass2-PLC* and opens at 1200×760 pixels in a horizontal split layout.

Toolbar. A top toolbar provides three buttons — *Open Script*, *Run*, and *Clear Output* — and a right-aligned status label that reports *Ready*, a loaded filename, or *Run failed*.

Left panel (source editor). A monospaced `Text` widget with both horizontal and vertical scrollbars allows the user to enter or paste arbitrary source code. The *Open Script* button

populates the editor from a file. The editor is pre-loaded with the built-in sample program on first launch.

Right panel (output tabs). A `Notebook` widget presents four tabs, each backed by a scrollable read-only `Text` widget with both horizontal and vertical scrollbars:

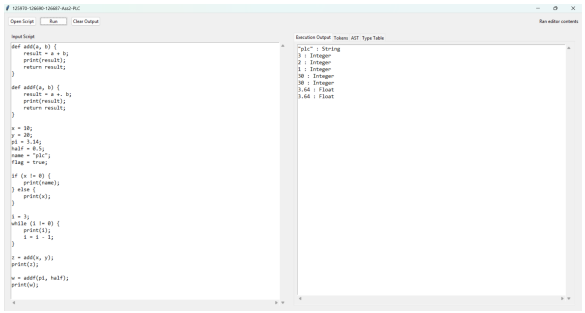
- *Execution Output* — the `print()` values produced by stage 4, formatted as `value : Type`;
- *Tokens* — the token stream from stage 1, one token per line with type, lexeme, and source position;
- *AST* — the indented tree text from stage 2;
- *Type Table* — the symbol table from stage 3 showing each name, kind, inferred type, initialisation state, and parameter list.

Error handling. If any pipeline stage raises an exception, the error message is written to the *Execution Output* tab as `ERROR: <message>`. No modal dialogs are used, so the user can edit and rerun without dismissing a popup.

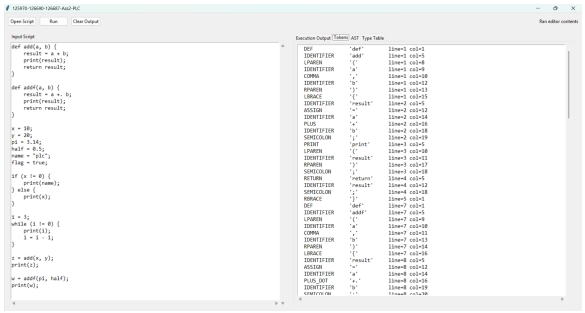
Keyboard shortcuts. `F5` and `Ctrl+Enter` are both bound to the *Run* action, allowing rapid edit-run cycles without reaching for the mouse.

6.6.1 Screenshots

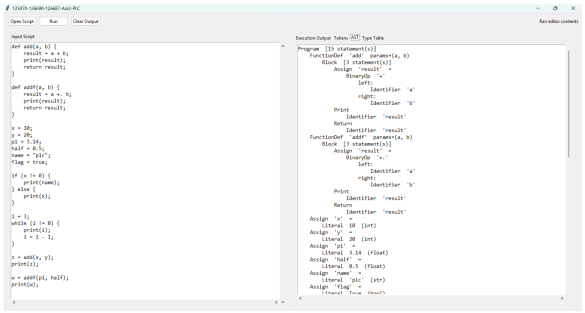
Figures 6.2 and 6.3 show the desktop workbench and CLI output.



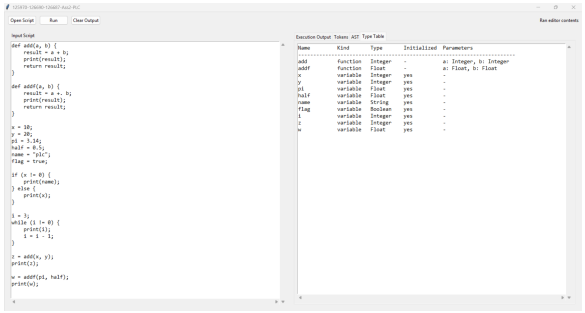
(a) Execution Output tab.



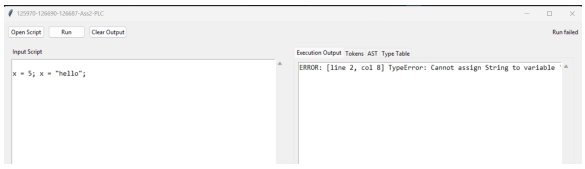
(b) Tokens tab (Stage 1).



(c) AST tab (Stage 2).



(d) Type Table tab (Stage 3).

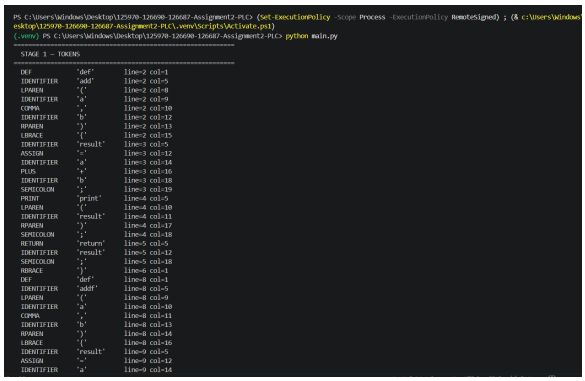


(e) Inline type-error display.

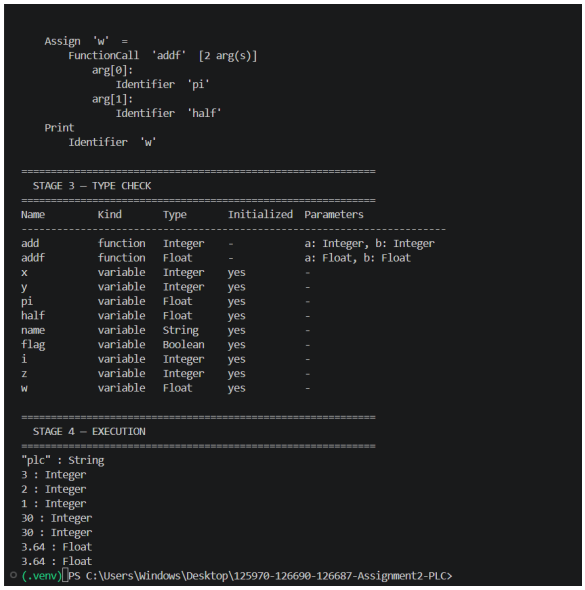
Figure 6.2. Desktop workbench — output tabs and error display.

6.7 Command-line interface

The CLI runner (`main.py`) accepts an optional script path; without one it runs the built-in sample and prints four labelled stages to `stdout`.



(a) Stage 1: token stream.



(b) Stage 3 type table and Stage 4 output.

Figure 6.3. CLI run of `python main.py` on the built-in sample.

CHAPTER 7

TESTING AND VERIFICATION

7.1 Manual Testing

Prior to the construction of the automated suite, each language feature was exercised manually by running the full pipeline and inspecting stage outputs. Table 7.1 summarises the categories covered.

Table 7.1. Representative manually tested categories

Category	Program Behaviour	Expected Result
Arithmetic	Evaluate mixed numeric expressions	Correct value and inferred type
If-then-else	Choose one branch from a Boolean condition	Only matching branch runs
While-loop	Print during repeated execution	One output line per iteration
Functions	Infer parameters and return type	Valid call and correct return value
Type mismatch	Reassign incompatible type	Type error before runtime

7.1.1 Lexer and Symbol Table

The lexer was verified manually by running the full pipeline and inspecting Stage 1 token output for programs that exercise every token category: integer and float literals, Boolean literals, string literals, all six keywords, two-character comparison operators (`==`, `!=`), single-character operators, and delimiters. Identifier scanning with keyword fallback was confirmed by checking that `true`, `false`, and reserved words are emitted with their keyword token types while non-keyword names produce `IDENTIFIER` tokens. For example, the source fragment `x = 10;` produces the token sequence:

```
IDENTIFIER  'x'    line=1 col=1
ASSIGN      '='    line=1 col=3
INT_LITERAL '10'   line=1 col=5
SEMICOLON   ';'    line=1 col=7
```

Error paths were also checked: supplying a bare `!` character raises a `LexerError` with a source

position and a message suggesting `!=`, and an unterminated string literal raises a `LexerError` citing the opening line.

The symbol table was verified through the type checker and the Stage 3 output. Duplicate variable definitions were confirmed to raise a `SymbolTableError` before execution. Scope isolation was verified with nested blocks: a variable defined inside an `if` or `while` body is not visible in the outer scope after the block ends. The `format_table()` output was checked against the expected column layout (name, kind, type, initialisation status, parameters) for both variable and function symbols.

7.1.2 Parser and AST

The parser was verified manually by running the full pipeline and inspecting Stage 2 AST output. The following cases were checked:

- arithmetic expressions with mixed precedence (e.g. `x + y * 2`) produce a tree where multiplication is a deeper sub-tree than addition, confirming the `_expr/_term/_factor` precedence encoding,
- assignment versus expression statement dispatch: `x = 5;` produces an `Assign` node while `add(1,2);` produces a `FunctionCall` node at the statement level, confirming the IDENTIFIER+ASSIGN lookahead,
- `if` with and without an `else` clause: the resulting `If` node has a non-null `else_block` only when the clause is present,
- function definitions store only parameter names in the `FunctionDef` node; the AST printer output was inspected to confirm no type annotations are present at this stage,
- string literals in `Literal` nodes have their surrounding double-quotes stripped; `print("hello")` prints `"hello" : String` at runtime (the interpreter re-adds the quotes when formatting string output),
- parse errors: missing semicolons, unmatched parentheses, and unexpected tokens all produce `[line X, col Y] ParseError:` messages with correct source positions.

7.1.3 Arithmetic Expressions

Arithmetic expressions were tested with integer-only and float-only programs to verify operator precedence and strict type separation. The expression `x + y * 2` confirmed that multiplication binds more tightly than addition, and mixing integer and float operands (e.g. `Integer + Float`) correctly produced a type error, confirming the no-overloading design.

7.1.4 Boolean Expressions

Boolean expressions were tested inside `if` and `while` conditions. The type checker correctly enforced that both operands of `==` and `!=` must be arithmetic; supplying a `Boolean` or `String` operand produced a source-positioned `TypeCheckError` before any execution.

7.1.5 Assignment Statements

Assignments were tested for both first-use type inference and reassignment consistency. Assigning `5` to a variable and subsequently assigning `"hello"` to the same variable correctly produced a type error, confirming that the inferred type is fixed after first assignment.

7.1.6 If-Then-Else

Conditional programs were constructed to verify that exactly one branch executes for a given condition and that neither branch is entered when the condition evaluates to the opposite truth value. Non-Boolean conditions were also supplied and confirmed to be rejected by the type checker before execution.

7.1.7 While-Loop

While-loops were tested with an integer countdown from three to zero. The results confirmed that the loop body executes once per iteration and terminates when the condition becomes false, with `print()` producing one output line per pass rather than accumulating output until loop exit.

7.1.8 Functions

Function tests covered value parameter passing, local scope execution, and return value propagation. The type checker correctly inferred parameter types from the first call site and enforced that signature for all subsequent calls to the same function.

7.1.9 Print

The built-in `print()` function was tested with all supported runtime types. The output includes the printed value together with its type, such as `3.14 : Float` and `"plc" : String`.

7.1.10 Unary Minus

The unary minus operator was verified for integer literals (`x = -5`), float literals (`y = -3.14`), negated variable references (`y = -x`), and negation inside a larger expression (`x = 3 + -2`). In each case the type checker assigned the correct type and the interpreter produced the expected value.

7.2 Type Error Detection

The checker was verified with invalid programs involving incompatible assignment, invalid arithmetic operands, wrong function arguments, and non-Boolean conditions. In each case the system stopped before execution and reported a source-positioned type error. Representative error messages are shown below.

```
# Assignment type mismatch
[line 2, col 1] TypeError: Cannot assign String to variable 'x' of type
    Integer

# Non-Boolean if condition
[line 1, col 4] TypeError: If condition must have type Boolean

# Wrong argument type for function
[line 5, col 9] TypeError: Argument 1 of 'add' must be Integer, got
   String
```

7.3 Automated Test Suite

The automated regression suite is located in `tests/test_pipeline.py` and uses Python's built-in `unittest` framework. The tests are organised into five classes, each targeting a distinct compiler stage.

Table 7.2. Automated test coverage by class

Test class	Stage covered	Tests
`LexerTests`	Tokenisation (`lexica.py`)	12
`SymbolTableTests`	Scoped symbol table (`symbol_table.py`)	7
`ParserTests`	Parsing and precedence (`parser.py`)	5
`TypeCheckerTests`	Static type inference (`type_checker.py`)	18
`IntegrationTests`	Full pipeline execution	17
	Total	**59**

LexerTests verify that each token category (integer, float, Boolean, string, keywords, identifiers, two-character operators) is produced correctly, that source line numbers are tracked across newlines, and that illegal characters and unterminated strings raise `LexerError`.

SymbolTableTests exercise the symbol table directly, without going through the full pipeline. They cover variable definition and lookup, duplicate definition raising `SymbolTableError` for both variables and functions, parent-scope lookup through `child_scope()`, undefined-symbol lookup error, and the `format_table()` output for both variable and function symbol rows.

TypeCheckerTests verify static type inference and error detection. The 18 tests cover integer and float arithmetic (including dot-suffixed operators), integer floor division, strict type separation (mixed Integer/Float rejected for both arithmetic and comparison operators), string and boolean type inference, assignment type mismatch, non-Boolean conditions, undefined variables, wrong argument count and type, and two tests for uncalled-function body checking: one confirms that a type error inside a never-called function body is detected, and another confirms that a valid uncalled function raises no error.

IntegrationTests run programs through all four pipeline stages and verify the final output. Cases include unary negation on all numeric types, both branches of `if/else`, while-loop countdown, `print()` with every type, value-parameter isolation, nested function calls, variable reassignment, and scope isolation (variables declared inside an `if` block are not visible outside it).

The tests can be run with:

```
python3 -m unittest discover -s tests -v
```

All 59 tests pass.

7.4 Error Handling

The language system reports errors at three main stages:

- lexical errors for malformed characters or unterminated strings,
- parse errors for invalid syntax such as missing semicolons,
- type errors for programs that violate static typing rules.

Runtime errors are also reported with source positions whenever execution reaches an invalid state such as an undefined variable reference.

CHAPTER 8

CONCLUSION

This project delivered a complete, statically-typed interpreted language supporting Integer, Float, Boolean, and String types, with arithmetic, comparisons, assignments, if-then-else, while-loops, functions with value parameter passing, and a built-in `print()` function.

The system follows a four-stage pipeline: lexer, parser, type checker, and interpreter. Variable and function types are inferred without explicit declarations; programs that violate type rules are rejected before execution. For uncalled functions, the type checker performs a post-pass to infer parameter types from body usage, ensuring type errors are always caught.

The 59 automated tests across five classes — lexical analysis, symbol table, parsing, type checking, and full pipeline — confirm correct behaviour for all language features. Both a command-line runner and a desktop UI are provided.

The implementation demonstrates all core compiler-construction concepts covered in the course — lexical analysis, parsing, static type inference, and tree-walking interpretation — in a complete, tested, and working system. Future work could extend the language with recursive functions, richer data structures, and code generation.

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APPENDIX A: SOURCE CODE

The complete source code is available on GitHub:

https://github.com/The-Token-Trio/125970-126690-126687-Assignment2-PLC

The project can be run from the command line or the desktop UI:

```
python3 main.py
python3 ui.py
python3 -m unittest discover -s tests -v
```

No external Python dependencies are required.

APPENDIX B: TEAM MEMBERS AND CONTRIBUTIONS

Name	Student ID	Contributions
Aye Khin Khin Hpone (Yolanda Lim)	st125970	Static typing rules; type checking; assignment, if, while, function, and `print()` execution; unary minus inference; pipeline integration (`pipeline.py`); automated test suite (59 tests)
Applegate T. Tun Oo	st126690	Lexer implementation (scanning, tokenisation, line/column tracking, error reporting); token definitions; keywords, operators, identifiers, and literals; symbol table (variable/function/type storage)
Win Htut Naing	st126687	Arithmetic and boolean expressions; assignment, if-then-else, while-loop, function definition/call, and `print()` parsing; unary minus parsing

A.1 Entry Points

```
from __future__ import annotations

import argparse
from pathlib import Path

from components.pipeline import format_stage_output, run_pipeline


```

```
DEFAULT_SOURCE = """
def add(a, b) {
    result = a + b;
    print(result);
    return result;
}

def addf(a, b) {
    result = a +. b;
    print(result);
    return result;
}

x = 10;
y = 20;
pi = 3.14;
half = 0.5;
name = "plc";
flag = true;

if (x != 0) {
    print(name);
} else {
    print(x);
}

i = 3;
while (i != 0) {
    print(i);
    i = i - 1;
}

z = add(x, y);
print(z);

w = addf(pi, half);
print(w);
"""


def run_source(source_code: str) -> None:
    result = run_pipeline(source_code)
    print(format_stage_output(result))


def _parse_args() -> argparse.Namespace:
    parser = argparse.ArgumentParser(description="Run the programming language pipeline.")
    parser.add_argument("script", nargs="?", help="Optional path to a source file to run")
    return parser.parse_args()


```

```
def main() -> None:
    args = _parse_args()
    if args.script:
        source_code = Path(args.script).read_text(encoding="utf-8")
    else:
        source_code = DEFAULT_SOURCE
    run_source(source_code)


if __name__ == "__main__":
    main()
```

Listing 8.1: main.py — command-line entry point

```
from __future__ import annotations

import tkinter as tk
from pathlib import Path
from tkinter import filedialog, ttk

from components.pipeline import format_tokens, run_pipeline


DEFAULT_SOURCE = """def add(a, b) {
    result = a + b;
    print(result);
    return result;
}

def addf(a, b) {
    result = a +. b;
    print(result);
    return result;
}

x = 10;
y = 20;
pi = 3.14;
half = 0.5;
name = \"plc\";
flag = true;

if (x != 0) {
    print(name);
} else {
    print(x);
}

i = 3;
while (i != 0) {
    print(i);
    i = i - 1;
}
```

```
z = add(x, y);
print(z);

w = addf(pi, half);
print(w);
"""


class LanguageWorkbench:
    def __init__(self) -> None:
        self.root = tk.Tk()
        self.root.title("125970-126690-126687-Ass2-PLC")
        self.root.geometry("1200x760")
        self.current_file: Path | None = None

        self._build_layout()
        self.source_text.insert("1.0", DEFAULT_SOURCE)
        self.root.bind("<F5>", lambda _e: self._run_source())
        self.root.bind("<Control-Return>", lambda _e: self._run_source
            ())

    def _build_layout(self) -> None:
        toolbar = ttk.Frame(self.root, padding=10)
        toolbar.pack(fill=tk.X)

        ttk.Button(toolbar, text="Open Script", command=self.
            _open_script).pack(side=tk.LEFT)
        ttk.Button(toolbar, text="Run", command=self._run_source).pack(
            side=tk.LEFT, padx=(8, 0))
        ttk.Button(toolbar, text="Clear Output", command=self.
            _clear_output).pack(side=tk.LEFT, padx=(8, 0))

        self.status_var = tk.StringVar(value="Ready")
        ttk.Label(toolbar, textvariable=self.status_var).pack(side=tk.
            RIGHT)

        body = ttk.Panedwindow(self.root, orient=tk.HORIZONTAL)
        body.pack(fill=tk.BOTH, expand=True, padx=10, pady=(0, 10))

        left_panel = ttk.Frame(body, padding=8)
        right_panel = ttk.Frame(body, padding=8)
        body.add(left_panel, weight=1)
        body.add(right_panel, weight=1)

        ttk.Label(left_panel, text="Input Script").pack(anchor=tk.W)
        src_frame = ttk.Frame(left_panel)
        src_frame.pack(fill=tk.BOTH, expand=True, pady=(6, 0))
        self.source_text = tk.Text(src_frame, wrap=tk.NONE, font=("
            Consolas", 11))
        src_vsb = ttk.Scrollbar(src_frame, orient=tk.VERTICAL, command=
            self.source_text.yview)
        src_hsb = ttk.Scrollbar(src_frame, orient=tk.HORIZONTAL,
```

```
            command=self.source_text.xview)
        self.source_text.configure(yscrollcommand=src_vsb.set,
            xscrollcommand=src_hsb.set)
        src_vsb.pack(side=tk.RIGHT, fill=tk.Y)
        src_hsb.pack(side=tk.BOTTOM, fill=tk.X)
        self.source_text.pack(side=tk.LEFT, fill=tk.BOTH, expand=True)

        notebook = ttk.Notebook(right_panel)
        notebook.pack(fill=tk.BOTH, expand=True)

        self.output_text = self._make_output_tab(notebook, "Execution
            Output")
        self.tokens_text = self._make_output_tab(notebook, "Tokens")
        self.ast_text = self._make_output_tab(notebook, "AST")
        self.types_text = self._make_output_tab(notebook, "Type Table")

    def _make_output_tab(self, notebook: ttk.Notebook, title: str) ->
        tk.Text:
        frame = ttk.Frame(notebook, padding=8)
        notebook.add(frame, text=title)
        text = tk.Text(frame, wrap=tk.NONE, font=("Consolas", 11),
            state=tk.DISABLED)
        vsb = ttk.Scrollbar(frame, orient=tk.VERTICAL, command=text.
            yview)
        hsb = ttk.Scrollbar(frame, orient=tk.HORIZONTAL, command=text.
            xview)
        text.configure(yscrollcommand=vsb.set, xscrollcommand=hsb.set)
        vsb.pack(side=tk.RIGHT, fill=tk.Y)
        hsb.pack(side=tk.BOTTOM, fill=tk.X)
        text.pack(side=tk.LEFT, fill=tk.BOTH, expand=True)
        return text

    def _open_script(self) -> None:
        file_path = filedialog.askopenfilename(
            title="Open Script File",
            filetypes=[("Text Files", "*.txt *.pl *.src"), ("All Files"
                , "*.*")],
        )
        if not file_path:
            return

        path = Path(file_path)
        self.current_file = path
        self.source_text.delete("1.0", tk.END)
        self.source_text.insert("1.0", path.read_text(encoding="utf-8")
            )
        self.status_var.set(f"Loaded {path.name}")

    def _run_source(self) -> None:
        source = self.source_text.get("1.0", tk.END)
        try:
            result = run_pipeline(source)
        except Exception as error:
```

```
            self._write_text(self.output_text, f"ERROR: {error}")
            self.status_var.set("Run failed")
            return

        self._write_text(self.output_text, "\n".join(result.outputs) if result.outputs else "(no output)")
        self._write_text(self.tokens_text, format_tokens(result.tokens))
        self._write_text(self.ast_text, result.ast_text)
        self._write_text(self.types_text, result.checked_scope.format_table())

        current_label = self.current_file.name if self.current_file is not None else "editor contents"
        self.status_var.set(f"Ran {current_label}")

    def _clear_output(self) -> None:
        self._write_text(self.output_text, "")
        self._write_text(self.tokens_text, "")
        self._write_text(self.ast_text, "")
        self._write_text(self.types_text, "")
        self.status_var.set("Output cleared")

    @staticmethod
    def _write_text(widget: tk.Text, content: str) -> None:
        widget.config(state=tk.NORMAL)
        widget.delete("1.0", tk.END)
        widget.insert("1.0", content)
        widget.config(state=tk.DISABLED)

    def run(self) -> None:
        self.root.mainloop()


if __name__ == "__main__":
    LanguageWorkbench().run()
```

Listing 8.2: ui.py — Tkinter desktop workbench

A.2 Core Components

```
from __future__ import annotations

from dataclasses import dataclass
from enum import Enum, auto


class TokenType(Enum):
    # Literals
    INT_LITERAL = auto()
    FLOAT_LITERAL = auto()
    BOOL_LITERAL = auto()
```

```
    STRING_LITERAL = auto()

    # Identifier
    IDENTIFIER = auto()

    # Keywords
    IF = auto()
    ELSE = auto()
    WHILE = auto()
    DEF = auto()
    RETURN = auto()
    PRINT = auto()

    # Operators
    PLUS = auto()
    MINUS = auto()
    STAR = auto()
    SLASH = auto()
    PLUS_DOT = auto()     # +.  (float addition)
    MINUS_DOT = auto()    # -.  (float subtraction)
    STAR_DOT = auto()     # *.  (float multiplication)
    SLASH_DOT = auto()    # /.  (float division)
    EQUAL_EQUAL = auto()
    BANG_EQUAL = auto()
    ASSIGN = auto()

    # Delimiters
    LPAREN = auto()
    RPAREN = auto()
    LBRACE = auto()
    RBRACE = auto()
    COMMA = auto()
    SEMICOLON = auto()

    EOF = auto()


@dataclass(frozen=True)
class Token:
    token_type: TokenType
    lexeme: str
    line: int
    column: int
```

Listing 8.3: components/tokens.py — token type enumeration and Token dataclass

```
from __future__ import annotations

from typing import Iterator

from components.tokens import Token, TokenType


```

```
KEYWORDS: dict[str, TokenType] = {
    "if": TokenType.IF,
    "else": TokenType.ELSE,
    "while": TokenType.WHILE,
    "def": TokenType.DEF,
    "return": TokenType.RETURN,
    "print": TokenType.PRINT,
    "true": TokenType.BOOL_LITERAL,
    "false": TokenType.BOOL_LITERAL,
}


# Arithmetic operators that may have a trailing '.' for the float variant
_ARITH_DOT: dict[str, tuple[TokenType, TokenType]] = {
    "+": (TokenType.PLUS,      TokenType.PLUS_DOT),
    "-": (TokenType.MINUS,     TokenType.MINUS_DOT),
    "*": (TokenType.STAR,      TokenType.STAR_DOT),
    "/": (TokenType.SLASH,     TokenType.SLASH_DOT),
}

SINGLE_CHAR_TOKENS: dict[str, TokenType] = {
    "=": TokenType.ASSIGN,
    "(": TokenType.LPAREN,
    ")": TokenType.RPAREN,
    "{": TokenType.LBRACE,
    "}": TokenType.RBRACE,
    ",": TokenType.COMMA,
    ";": TokenType.SEMICOLON,
}


class LexerError(Exception):
    pass


class Lexer:
    def __init__(self, source: str) -> None:
        self.source = source
        self.start = 0
        self.current = 0
        self.line = 1
        self.column = 1
        self.token_column = 1

    def tokenize(self) -> list[Token]:
        tokens: list[Token] = []
        while not self._is_at_end():
            self.start = self.current
            self.token_column = self.column
            token = self._scan_token()
            if token is not None:
                tokens.append(token)
```

```
        tokens.append(Token(TokenType.EOF, "", self.line, self.column))
        return tokens

    def _scan_token(self) -> Token | None:
        char = self._advance()

        # Whitespace handling
        if char in {" ", "\t", "\r"}:
            return None
        if char == "\n":
            self.line += 1
            self.column = 1
            return None

        # Two-char operators
        if char == "=":
            if self._match("="):
                return self._make_token(TokenType.EQUAL_EQUAL)
            return self._make_token(TokenType.ASSIGN)
        if char == "!":
            if self._match("="):
                return self._make_token(TokenType.BANG_EQUAL)
            raise LexerError(self._error_message("Unexpected '!'. Did
                you mean '!='?"))

        # Arithmetic operators: + - * /  (with optional trailing '.'
            for float variant)
        arith = _ARITH_DOT.get(char)
        if arith is not None:
            int_tok, float_tok = arith
            # Unary-minus lookahead: '-' followed by a digit is NOT
                '-.' even if
            # next char is '.' -- we still need to check char after '.'
                is not digit
            # (to avoid consuming the '.' of a float literal like 3.14)
                .
            # Rule: X. is a float operator only when the char after '.'
                 is NOT a digit.
            if self._peek() == "." and not self._peek_next().isdigit():
                self._advance()  # consume '.'
                return self._make_token(float_tok)
            return self._make_token(int_tok)

        # Single-char operators/delimiters
        single_char_token = SINGLE_CHAR_TOKENS.get(char)
        if single_char_token is not None:
            return self._make_token(single_char_token)

        if char == '"':
            return self._string()

        if char.isdigit():
            return self._number()
```

```
107
108         if self._is_identifier_start(char):
109             return self._identifier()
110
111         raise LexerError(self._error_message(f"Unexpected character: {
                char!r}"))
112
113     def _string(self) -> Token:
114         while self._peek() != '"' and not self._is_at_end():
115             if self._peek() == "\n":
116                 self.line += 1
117                 self.column = 1
118             self._advance()
119
120         if self._is_at_end():
121             raise LexerError(self._error_message("Unterminated string
                    literal"))
122
123         # Closing quote
124         self._advance()
125         return self._make_token(TokenType.STRING_LITERAL)
126
127     def _number(self) -> Token:
128         while self._peek().isdigit():
129             self._advance()
130
131         is_float = False
132         if self._peek() == "." and self._peek_next().isdigit():
133             is_float = True
134             self._advance()
135             while self._peek().isdigit():
136                 self._advance()
137
138         if is_float:
139             return self._make_token(TokenType.FLOAT_LITERAL)
140         return self._make_token(TokenType.INT_LITERAL)
141
142     def _identifier(self) -> Token:
143         while self._is_identifier_part(self._peek()):
144             self._advance()
145
146         lexeme = self._current_lexeme()
147         token_type = KEYWORDS.get(lexeme, TokenType.IDENTIFIER)
148         return self._make_token(token_type)
149
150     def _current_lexeme(self) -> str:
151         return self.source[self.start : self.current]
152
153     def _make_token(self, token_type: TokenType) -> Token:
154         return Token(token_type, self._current_lexeme(), self.line,
                self.token_column)
155
156     def _advance(self) -> str:
```

```
        char = self.source[self.current]
        self.current += 1
        self.column += 1
        return char

    def _match(self, expected: str) -> bool:
        if self._is_at_end():
            return False
        if self.source[self.current] != expected:
            return False

        self.current += 1
        self.column += 1
        return True

    def _peek(self) -> str:
        if self._is_at_end():
            return "\0"
        return self.source[self.current]

    def _peek_next(self) -> str:
        if self.current + 1 >= len(self.source):
            return "\0"
        return self.source[self.current + 1]

    def _is_at_end(self) -> bool:
        return self.current >= len(self.source)

    @staticmethod
    def _is_identifier_start(char: str) -> bool:
        return char.isalpha() or char == "_"

    @staticmethod
    def _is_identifier_part(char: str) -> bool:
        return char.isalnum() or char == "_"

    def _error_message(self, message: str) -> str:
        return f"[line {self.line}, col {self.token_column}] {message}"


def iter_tokens(source: str) -> Iterator[Token]:
    yield from Lexer(source).tokenize()
```

Listing 8.4: components/lexica.py — lexical analyser

```
from __future__ import annotations

from dataclasses import dataclass, field
from typing import Optional


#
    ------------------------------------------------------------------------
```

```python

# Base
#
    ------------------------------------------------------------------------


@dataclass
class ASTNode:
    """Base class for every AST node. Carries source location for
       diagnostics."""
    line: int = field(default=0, kw_only=True)
    column: int = field(default=0, kw_only=True)


#
    ------------------------------------------------------------------------

# Expressions
#
    ------------------------------------------------------------------------


@dataclass
class Literal(ASTNode):
    """An integer, float, boolean, or string literal.

    Examples:  42   3.14   true   "hello"
    """
    value: int | float | bool | str


@dataclass
class Identifier(ASTNode):
    """A variable or function name used as an expression.

    Example:  x
    """
    name: str


@dataclass
class BinaryOp(ASTNode):
    """An arithmetic or comparison binary operation.

    op is one of: '+' '-' '*' '/' '+.' '-.' '*.' '/.' '==' '!='
    Example:  a + b   x == 0   a +. b
    """
    op: str
    left: ASTNode
    right: ASTNode


@dataclass
```

```
class FunctionCall(ASTNode):
    """A call to a user-defined function.

    Example:  add(1, 2)
    """
    name: str
    args: list[ASTNode] = field(default_factory=list)


#
    ------------------------------------------------------------------------

# Statements
#
    ------------------------------------------------------------------------


@dataclass
class Assign(ASTNode):
    """Variable assignment (first use infers type; later uses must
       match).

    Example:  x = 10;
    """
    name: str
    value: ASTNode


@dataclass
class Print(ASTNode):
    """Built-in print statement.

    Example:  print(x);
    """
    expr: ASTNode


@dataclass
class Return(ASTNode):
    """Return statement inside a function body.

    Example:  return result;
    """
    expr: ASTNode


@dataclass
class Block(ASTNode):
    """A brace-enclosed sequence of statements.

    Example:  { x = 1; print(x); }
    """
    statements: list[ASTNode] = field(default_factory=list)
```

```
101
102
103 @dataclass
104 class If(ASTNode):
105     """If / optional else control flow.
106
107     Example:  if (x != 0) { print(x); } else { print(0); }
108     """
109     condition: ASTNode
110     then_block: Block
111     else_block: Optional[Block]
112
113
114 @dataclass
115 class While(ASTNode):
116     """While loop.
117
118     Example:  while (x != 0) { x = x - 1; }
119     """
120     condition: ASTNode
121     body: Block
122
123
124 @dataclass
125 class FunctionDef(ASTNode):
126     """Function definition.
127
128     Example:  def add(a, b) { return a + b; }
129
130     Note: parameter types are unknown at parse time -- Member 3's type
        checker
131     will infer them from usage. We store only the parameter names here.
132     """
133     name: str
134     params: list[str]
135     body: Block
136
137
138 #
    ------------------------------------------------------------------------

139 # Top-level
140 #
    ------------------------------------------------------------------------

141
142 @dataclass
143 class Program(ASTNode):
144     """Root node: the entire source file as a list of statements."""
145     statements: list[ASTNode] = field(default_factory=list)
```

Listing 8.5: components/ast_nodes.py — AST node definitions

```python
from __future__ import annotations

from typing import Optional

from components.tokens import Token, TokenType
from components.ast_nodes import (
    ASTNode,
    Program,
    Block,
    Assign,
    If,
    While,
    FunctionDef,
    FunctionCall,
    Print,
    Return,
    BinaryOp,
    Literal,
    Identifier,
)


# ---------------------------------------------------------------------------
# Parser error
# ---------------------------------------------------------------------------


class ParseError(Exception):
    pass


# ---------------------------------------------------------------------------
# Parser
# ---------------------------------------------------------------------------


class Parser:
    """Recursive-descent parser.

    Usage:
        tokens = Lexer(source).tokenize()
        tree   = Parser(tokens).parse()   # returns Program node
    """

    def __init__(self, tokens: list[Token]) -> None:
        self._tokens = tokens
```

```
        self._pos = 0

    #
        ------------------------------------------------------------------------

    # Public entry point
    #
        ------------------------------------------------------------------------


    def parse(self) -> Program:
        """Parse the full token stream and return the root Program node
            ."""
        line, col = self._peek().line, self._peek().column
        statements: list[ASTNode] = []
        while not self._is_at_end():
            statements.append(self._statement())
        return Program(statements=statements, line=line, column=col)

    #
        ------------------------------------------------------------------------

    # Statements
    #
        ------------------------------------------------------------------------


    def _statement(self) -> ASTNode:
        """Dispatch to the correct statement parser based on the
           current token."""
        tok = self._peek()

        if tok.token_type == TokenType.DEF:
            return self._func_def()

        if tok.token_type == TokenType.IF:
            return self._if_stmt()

        if tok.token_type == TokenType.WHILE:
            return self._while_stmt()

        if tok.token_type == TokenType.RETURN:
            return self._return_stmt()

        if tok.token_type == TokenType.PRINT:
            return self._print_stmt()

        # assignment: IDENTIFIER "=" expr ";"
        if (
            tok.token_type == TokenType.IDENTIFIER
            and self._peek_next().token_type == TokenType.ASSIGN
        ):
            return self._assignment()
```

```
88
89          # fallback: bare expression statement  expr ";"
90          node = self._expr()
91          self._expect(TokenType.SEMICOLON, "Expected ';' after
                expression")
92          return node
93
94      def _assignment(self) -> Assign:
95          """IDENTIFIER '=' expr ';'"""
96          name_tok = self._advance()                              #
                IDENTIFIER
97          self._advance()                                         # '='
98          value = self._expr()
99          self._expect(TokenType.SEMICOLON, "Expected ';' after
                assignment")
100         return Assign(name=name_tok.lexeme, value=value,
101                       line=name_tok.line, column=name_tok.column)
102
103     def _if_stmt(self) -> If:
104         """'if' '(' bool_expr ')' block ( 'else' block )?"""
105         tok = self._advance()                                   # 'if'
106         self._expect(TokenType.LPAREN, "Expected '(' after 'if'")
107         condition = self._bool_expr()
108         self._expect(TokenType.RPAREN, "Expected ')' after if condition
                ")
109         then_block = self._block()
110         else_block: Optional[Block] = None
111         if self._check(TokenType.ELSE):
112             self._advance()                                     # 'else
                    '
113             else_block = self._block()
114         return If(condition=condition, then_block=then_block,
                else_block=else_block,
115                   line=tok.line, column=tok.column)
116
117     def _while_stmt(self) -> While:
118         """'while' '(' bool_expr ')' block"""
119         tok = self._advance()                                   # '
                while'
120         self._expect(TokenType.LPAREN, "Expected '(' after 'while'")
121         condition = self._bool_expr()
122         self._expect(TokenType.RPAREN, "Expected ')' after while
                condition")
123         body = self._block()
124         return While(condition=condition, body=body,
125                      line=tok.line, column=tok.column)
126
127     def _func_def(self) -> FunctionDef:
128         """'def' IDENTIFIER '(' params? ')' block"""
129         tok = self._advance()                                   # 'def'
130         name_tok = self._expect(TokenType.IDENTIFIER, "Expected
                function name after 'def'")
131         self._expect(TokenType.LPAREN, "Expected '(' after function
```

```
            name")
132         params = self._params()
133         self._expect(TokenType.RPAREN, "Expected ')' after parameters")
134         body = self._block()
135         return FunctionDef(name=name_tok.lexeme, params=params, body=
                body,
136                            line=tok.line, column=tok.column)
137
138     def _print_stmt(self) -> Print:
139         """'print' '(' expr ')' ';'"""
140         tok = self._advance()                                    # '
                print'
141         self._expect(TokenType.LPAREN, "Expected '(' after 'print'")
142         expr = self._expr()
143         self._expect(TokenType.RPAREN, "Expected ')' after print
                expression")
144         self._expect(TokenType.SEMICOLON, "Expected ';' after print
                statement")
145         return Print(expr=expr, line=tok.line, column=tok.column)
146
147     def _return_stmt(self) -> Return:
148         """'return' expr ';'"""
149         tok = self._advance()                                    # '
                return'
150         expr = self._expr()
151         self._expect(TokenType.SEMICOLON, "Expected ';' after return
                value")
152         return Return(expr=expr, line=tok.line, column=tok.column)
153
154     def _block(self) -> Block:
155         """'{' statement* '}'"""
156         tok = self._expect(TokenType.LBRACE, "Expected '{'")
157         statements: list[ASTNode] = []
158         while not self._check(TokenType.RBRACE) and not self._is_at_end
                ():
159             statements.append(self._statement())
160         self._expect(TokenType.RBRACE, "Expected '}' to close block")
161         return Block(statements=statements, line=tok.line, column=tok.
                column)
162
163     #
        ------------------------------------------------------------------------

164     # Parameters (function definition)
165     #
        ------------------------------------------------------------------------

166
167     def _params(self) -> list[str]:
168         """IDENTIFIER (',' IDENTIFIER)*  -- returns list of param names.
                """
169         params: list[str] = []
170         if self._check(TokenType.IDENTIFIER):
```

```
171             params.append(self._advance().lexeme)
172             while self._check(TokenType.COMMA):
173                 self._advance()                                    # ','
174                 tok = self._expect(TokenType.IDENTIFIER, "Expected
                        parameter name after ','")
175                 params.append(tok.lexeme)
176         return params
177
178     #
        ------------------------------------------------------------------------

179     # Boolean expressions  (lowest precedence, used in if/while
        conditions)
180     #
        ------------------------------------------------------------------------

181
182     def _bool_expr(self) -> ASTNode:
183         """expr ('==' | '!=') expr   |   BOOL_LITERAL"""
184         # bare boolean literal: true / false
185         if self._check(TokenType.BOOL_LITERAL):
186             tok = self._advance()
187             value = tok.lexeme == "true"
188             return Literal(value=value, line=tok.line, column=tok.
                    column)
189
190         left = self._expr()
191
192         if self._check(TokenType.EQUAL_EQUAL):
193             op_tok = self._advance()
194             right = self._expr()
195             return BinaryOp(op="==", left=left, right=right,
196                             line=op_tok.line, column=op_tok.column)
197
198         if self._check(TokenType.BANG_EQUAL):
199             op_tok = self._advance()
200             right = self._expr()
201             return BinaryOp(op="!=", left=left, right=right,
202                             line=op_tok.line, column=op_tok.column)
203
204         # no comparison operator found -- return the expr as-is
205         # (the type checker will validate it's Boolean)
206         return left
207
208     #
        ------------------------------------------------------------------------

209     # Arithmetic expressions  (standard precedence)
210     #
        ------------------------------------------------------------------------

211
212     def _expr(self) -> ASTNode:
```

```
        """term (('+' | '-' | '+.' | '-.') term)*"""
        node = self._term()
        while self._check(TokenType.PLUS) or self._check(TokenType.
            MINUS) \
                or self._check(TokenType.PLUS_DOT) or self._check(
                    TokenType.MINUS_DOT):
            op_tok = self._advance()
            right = self._term()
            node = BinaryOp(op=op_tok.lexeme, left=node, right=right,
                            line=op_tok.line, column=op_tok.column)
        return node

    def _term(self) -> ASTNode:
        """factor (('*' | '/' | '*.' | '/.') factor)*"""
        node = self._factor()
        while self._check(TokenType.STAR) or self._check(TokenType.
            SLASH) \
                or self._check(TokenType.STAR_DOT) or self._check(
                    TokenType.SLASH_DOT):
            op_tok = self._advance()
            right = self._factor()
            node = BinaryOp(op=op_tok.lexeme, left=node, right=right,
                            line=op_tok.line, column=op_tok.column)
        return node

    def _factor(self) -> ASTNode:
        """
        '-' factor  (unary negation, desugared to 0 - factor)
        | INT_LITERAL | FLOAT_LITERAL | STRING_LITERAL | BOOL_LITERAL
        | IDENTIFIER ( '(' args? ')' )?
        | '(' expr ')'
        """
        tok = self._peek()

        # Unary minus  '-' factor  (integer negation, desugared to 0 -
            factor)
        if tok.token_type == TokenType.MINUS:
            op_tok = self._advance()
            operand = self._factor()
            zero = Literal(value=0, line=op_tok.line, column=op_tok.
                column)
            return BinaryOp(op="-", left=zero, right=operand,
                            line=op_tok.line, column=op_tok.column)

        # Unary minus-dot  '-.' factor  (float negation, desugared to
            0.0 -. factor)
        if tok.token_type == TokenType.MINUS_DOT:
            op_tok = self._advance()
            operand = self._factor()
            zero_f = Literal(value=0.0, line=op_tok.line, column=op_tok
                .column)
            return BinaryOp(op="-.", left=zero_f, right=operand,
                            line=op_tok.line, column=op_tok.column)
```

```
258
259         # Integer literal
260         if tok.token_type == TokenType.INT_LITERAL:
261             self._advance()
262             return Literal(value=int(tok.lexeme), line=tok.line, column
                    =tok.column)
263
264         # Float literal
265         if tok.token_type == TokenType.FLOAT_LITERAL:
266             self._advance()
267             return Literal(value=float(tok.lexeme), line=tok.line,
                    column=tok.column)
268
269         # Boolean literal
270         if tok.token_type == TokenType.BOOL_LITERAL:
271             self._advance()
272             return Literal(value=(tok.lexeme == "true"), line=tok.line,
                     column=tok.column)
273
274         # String literal  (keep the surrounding quotes stripped)
275         if tok.token_type == TokenType.STRING_LITERAL:
276             self._advance()
277             return Literal(value=tok.lexeme[1:-1], line=tok.line,
                    column=tok.column)
278
279         # Identifier -- could be a plain variable or a function call
280         if tok.token_type == TokenType.IDENTIFIER:
281             self._advance()
282             if self._check(TokenType.LPAREN):
283                 # function call
284                 self._advance()                                    # '('
285                 args = self._args()
286                 self._expect(TokenType.RPAREN, "Expected ')' after
                        function arguments")
287                 return FunctionCall(name=tok.lexeme, args=args,
288                                     line=tok.line, column=tok.column)
289             return Identifier(name=tok.lexeme, line=tok.line, column=
                    tok.column)
290
291         # Grouped expression  '(' expr ')'
292         if tok.token_type == TokenType.LPAREN:
293             self._advance()                                        # '('
294             node = self._expr()
295             self._expect(TokenType.RPAREN, "Expected ')' to close
                    grouped expression")
296             return node
297
298         raise ParseError(self._error_at(tok, f"Unexpected token: {tok.
                lexeme!r}"))
299
300     #
        ----------------------------------------------------------------------
```

```
    # Arguments (function call)
    #
        ------------------------------------------------------------------------

    def _args(self) -> list[ASTNode]:
        """expr (',' expr)*"""
        args: list[ASTNode] = []
        if not self._check(TokenType.RPAREN):
            args.append(self._expr())
            while self._check(TokenType.COMMA):
                self._advance()                                  # ','
                args.append(self._expr())
        return args

    #
        ------------------------------------------------------------------------

    # Token navigation helpers
    #
        ------------------------------------------------------------------------

    def _peek(self) -> Token:
        return self._tokens[self._pos]

    def _peek_next(self) -> Token:
        if self._pos + 1 < len(self._tokens):
            return self._tokens[self._pos + 1]
        return self._tokens[-1]                                  # EOF

    def _advance(self) -> Token:
        tok = self._tokens[self._pos]
        if not self._is_at_end():
            self._pos += 1
        return tok

    def _check(self, token_type: TokenType) -> bool:
        return self._peek().token_type == token_type

    def _is_at_end(self) -> bool:
        return self._peek().token_type == TokenType.EOF

    def _expect(self, token_type: TokenType, message: str) -> Token:
        if self._check(token_type):
            return self._advance()
        raise ParseError(self._error_at(self._peek(), message))

    def _error_at(self, tok: Token, message: str) -> str:
        return f"[line {tok.line}, col {tok.column}] ParseError: {
           message}"
```

Listing 8.6: components/parser.py — recursive-descent parser

```python
from __future__ import annotations

from dataclasses import dataclass, field
from enum import Enum


class LanguageType(Enum):
    INTEGER = "Integer"
    FLOAT = "Float"
    BOOLEAN = "Boolean"
    STRING = "String"
    VOID = "Void"


@dataclass
class Symbol:
    name: str
    symbol_type: LanguageType


@dataclass
class VariableSymbol(Symbol):
    initialized: bool = False


@dataclass
class FunctionSymbol(Symbol):
    parameters: list[tuple[str, LanguageType]] = field(default_factory=
        list)


class SymbolTableError(Exception):
    pass


class SymbolTable:
    def __init__(self, parent: SymbolTable | None = None) -> None:
        self.parent = parent
        self._symbols: dict[str, Symbol] = {}

    def define_variable(self, name: str, symbol_type: LanguageType,
        initialized: bool = False) -> VariableSymbol:
        if name in self._symbols:
            raise SymbolTableError(f"Symbol already defined in this
                scope: {name}")
        symbol = VariableSymbol(name=name, symbol_type=symbol_type,
            initialized=initialized)
        self._symbols[name] = symbol
        return symbol

    def define_function(
        self,
```

```
        name: str,
        return_type: LanguageType,
        parameters: list[tuple[str, LanguageType]],
    ) -> FunctionSymbol:
        if name in self._symbols:
            raise SymbolTableError(f"Symbol already defined in this scope: {name}")
        symbol = FunctionSymbol(name=name, symbol_type=return_type, parameters=parameters)
        self._symbols[name] = symbol
        return symbol

    def lookup(self, name: str) -> Symbol:
        symbol = self._symbols.get(name)
        if symbol is not None:
            return symbol
        if self.parent is not None:
            return self.parent.lookup(name)
        raise SymbolTableError(f"Undefined symbol: {name}")

    def exists_in_current_scope(self, name: str) -> bool:
        return name in self._symbols

    def child_scope(self) -> SymbolTable:
        return SymbolTable(parent=self)

    def set_initialized(self, name: str) -> None:
        symbol = self.lookup(name)
        if not isinstance(symbol, VariableSymbol):
            raise SymbolTableError(f"Cannot initialize non-variable symbol: {name}")
        symbol.initialized = True

    def snapshot(self) -> dict[str, Symbol]:
        return dict(self._symbols)

    def format_table(self) -> str:
        lines = [
            f"{'Name':<12} {'Kind':<10} {'Type':<10} {'Initialized':<12} Parameters",
            "-" * 72,
        ]

        for name, symbol in self._symbols.items():
            if isinstance(symbol, FunctionSymbol):
                params = ", ".join(f"{param_name}: {param_type.value}"
                    for param_name, param_type in symbol.parameters)
                lines.append(
                    f"{name:<12} {'function':<10} {symbol.symbol_type.value:<10} {'-':<12} {params or '-'}"
                )
            elif isinstance(symbol, VariableSymbol):
                initialized = "yes" if symbol.initialized else "no"
```

```
                    lines.append(
                        f"{name:<12} {'variable':<10} {symbol.symbol_type.
                            value:<10} {initialized:<12} -"
                    )
                else:
                    lines.append(f"{name:<12} {'symbol':<10} {symbol.
                        symbol_type.value:<10} {'-':<12} -")

        return "\n".join(lines)
```

Listing 8.7: components/symbol_table.py — scoped symbol table

```
from __future__ import annotations

from dataclasses import dataclass

from components.ast_nodes import (
    ASTNode,
    Assign,
    BinaryOp,
    Block,
    FunctionCall,
    FunctionDef,
    Identifier,
    If,
    Literal,
    Print,
    Program,
    Return,
    While,
)
from components.symbol_table import FunctionSymbol, LanguageType,
    SymbolTable, SymbolTableError, VariableSymbol


class TypeCheckError(Exception):
    pass


@dataclass
class _FunctionState:
    node: FunctionDef
    body_checked: bool = False


@dataclass
class _FunctionContext:
    name: str
    return_type: LanguageType | None = None


class TypeChecker:
    def __init__(self) -> None:
```

```
        self.global_scope = SymbolTable()
        self._functions: dict[str, _FunctionState] = {}
        self._active_calls: list[str] = []

    def check(self, program: Program) -> SymbolTable:
        for statement in program.statements:
            if isinstance(statement, FunctionDef):
                self._register_function(statement)

        for statement in program.statements:
            if isinstance(statement, FunctionDef):
                continue
            self._check_statement(statement, self.global_scope, None)

        self._check_uncalled_functions()
        return self.global_scope

    def _register_function(self, node: FunctionDef) -> None:
        if node.name in self._functions:
            raise TypeCheckError(self._error_at(node, f"Function '{node
                .name}' is already defined"))

        placeholder_parameters = [(param_name, LanguageType.VOID) for
            param_name in node.params]
        try:
            self.global_scope.define_function(node.name, LanguageType.
                VOID, placeholder_parameters)
        except SymbolTableError as error:
            raise TypeCheckError(self._error_at(node, str(error))) from
                error
        self._functions[node.name] = _FunctionState(node=node)

    def _check_statement(
        self,
        node: ASTNode,
        scope: SymbolTable,
        function_context: _FunctionContext | None,
    ) -> None:
        if isinstance(node, Assign):
            value_type = self._infer_expr_type(node.value, scope)
            self._assign_variable_type(scope, node, value_type)
            return

        if isinstance(node, Print):
            self._infer_expr_type(node.expr, scope)
            return

        if isinstance(node, Return):
            if function_context is None:
                raise TypeCheckError(self._error_at(node, "'return' is
                    only valid inside a function"))
            expr_type = self._infer_expr_type(node.expr, scope)
            if function_context.return_type is None:
```

```
                function_context.return_type = expr_type
            elif function_context.return_type != expr_type:
                raise TypeCheckError(
                    self._error_at(
                        node,
                        f"Function '{function_context.name}' returns
                            both {function_context.return_type.value}
                            and {expr_type.value}",
                    )
                )
            return

        if isinstance(node, If):
            condition_type = self._infer_expr_type(node.condition,
                scope)
            if condition_type != LanguageType.BOOLEAN:
                raise TypeCheckError(self._error_at(node.condition, "If
                    condition must have type Boolean"))
            self._check_block(node.then_block, scope.child_scope(),
                function_context)
            if node.else_block is not None:
                self._check_block(node.else_block, scope.child_scope(),
                    function_context)
            return

        if isinstance(node, While):
            condition_type = self._infer_expr_type(node.condition,
                scope)
            if condition_type != LanguageType.BOOLEAN:
                raise TypeCheckError(self._error_at(node.condition, "
                    While condition must have type Boolean"))
            self._check_block(node.body, scope.child_scope(),
                function_context)
            return

        if isinstance(node, Block):
            self._check_block(node, scope.child_scope(),
                function_context)
            return

        if isinstance(node, FunctionCall):
            self._infer_function_call_type(node, scope)
            return

        raise TypeCheckError(self._error_at(node, f"Unsupported
            statement node: {type(node).__name__}"))

    def _check_block(
        self,
        block: Block,
        scope: SymbolTable,
        function_context: _FunctionContext | None,
    ) -> None:
```

```
131             for statement in block.statements:
132                 self._check_statement(statement, scope, function_context)
133 
134     def _infer_expr_type(self, node: ASTNode, scope: SymbolTable) ->
        LanguageType:
135         if isinstance(node, Literal):
136             return self._literal_type(node)
137 
138         if isinstance(node, Identifier):
139             try:
140                 symbol = scope.lookup(node.name)
141             except SymbolTableError as error:
142                 raise TypeCheckError(self._error_at(node, str(error)))
                    from error
143             if not isinstance(symbol, VariableSymbol):
144                 raise TypeCheckError(self._error_at(node, f"'{node.name
                    }' is not a variable"))
145             return symbol.symbol_type
146 
147         if isinstance(node, FunctionCall):
148             return self._infer_function_call_type(node, scope)
149 
150         if isinstance(node, BinaryOp):
151             left_type = self._infer_expr_type(node.left, scope)
152             right_type = self._infer_expr_type(node.right, scope)
153             return self._infer_binary_type(node, left_type, right_type)
154 
155         raise TypeCheckError(self._error_at(node, f"Unsupported
            expression node: {type(node).__name__}"))
156 
157     def _infer_function_call_type(self, node: FunctionCall, scope:
        SymbolTable) -> LanguageType:
158         function_state = self._functions.get(node.name)
159         if function_state is None:
160             raise TypeCheckError(self._error_at(node, f"Undefined
                function: {node.name}"))
161         if node.name in self._active_calls:
162             raise TypeCheckError(self._error_at(node, f"Recursive
                function calls are not supported: {node.name}"))
163 
164         argument_types = [self._infer_expr_type(argument, scope) for
            argument in node.args]
165         function_symbol = self._lookup_function_symbol(node)
166 
167         if len(argument_types) != len(function_symbol.parameters):
168             raise TypeCheckError(
169                 self._error_at(
170                     node,
171                     f"Function '{node.name}' expects {len(
                        function_symbol.parameters)} argument(s), got {
                        len(argument_types)}",
172                 )
173             )
```

```
174
175         declared_parameter_types = [param_type for _, param_type in
                function_symbol.parameters]
176         if all(param_type == LanguageType.VOID for param_type in
                declared_parameter_types):
177             function_symbol.parameters = list(zip(function_state.node.
                    params, argument_types, strict=False))
178         else:
179             for index, ((_, expected_type), actual_type) in enumerate(
                    zip(function_symbol.parameters, argument_types, strict=
                    False)):
180                 if expected_type != actual_type:
181                     raise TypeCheckError(
182                         self._error_at(
183                             node.args[index],
184                             f"Argument {index + 1} of '{node.name}'
                                    must be {expected_type.value}, got {
                                    actual_type.value}",
185                         )
186                     )
187
188         if not function_state.body_checked:
189             self._check_function_body(function_state, function_symbol)
190
191         return function_symbol.symbol_type
192
193     def _check_function_body(self, function_state: _FunctionState,
            function_symbol: FunctionSymbol) -> None:
194         function_context = _FunctionContext(name=function_state.node.
                name)
195         function_scope = self.global_scope.child_scope()
196         for parameter_name, parameter_type in function_symbol.
                parameters:
197             function_scope.define_variable(parameter_name,
                    parameter_type, initialized=True)
198
199         self._active_calls.append(function_state.node.name)
200         try:
201             self._check_block(function_state.node.body, function_scope,
                        function_context)
202         finally:
203             self._active_calls.pop()
204
205         function_symbol.symbol_type = function_context.return_type or
                LanguageType.VOID
206         function_state.body_checked = True
207
208     def _check_uncalled_functions(self) -> None:
209         """After all call-site checks, type-check any function whose
                body was never visited."""
210         for name, state in self._functions.items():
211             if not state.body_checked:
212                 try:
```

```
                    function_symbol = self.global_scope.lookup(name)
                except SymbolTableError:
                    continue
                if not isinstance(function_symbol, FunctionSymbol):
                    continue
                inferred_params = self._infer_param_types_from_body(
                    state.node)
                function_symbol.parameters = inferred_params
                self._check_function_body(state, function_symbol)

    def _infer_param_types_from_body(self, node: FunctionDef) -> list[
        tuple[str, LanguageType]]:
        """Infer parameter types by scanning the body for expression
            contexts.

        Any parameter used as an operand of an arithmetic or comparison
            operator
        is inferred as numeric (Float if the sibling operand is a float
            literal,
        Integer otherwise).  Parameters whose usage gives no type hint
            default to
        Integer.
        """
        param_names = set(node.params)
        inferred: dict[str, LanguageType] = {}

        def scan_expr(n: ASTNode) -> None:
            if isinstance(n, BinaryOp):
                if n.op in {"+", "-", "*", "/", "==", "!="}:
                    for operand in (n.left, n.right):
                        if isinstance(operand, Identifier) and operand.
                            name in param_names:
                            inferred.setdefault(operand.name,
                                LanguageType.INTEGER)
                elif n.op in {"+.", "-.", "*.", "/."}:
                    for operand in (n.left, n.right):
                        if isinstance(operand, Identifier) and operand.
                            name in param_names:
                            inferred.setdefault(operand.name,
                                LanguageType.FLOAT)
                scan_expr(n.left)
                scan_expr(n.right)
            elif isinstance(n, FunctionCall):
                for arg in n.args:
                    scan_expr(arg)

        def scan_stmt(n: ASTNode) -> None:
            if isinstance(n, Assign):
                scan_expr(n.value)
            elif isinstance(n, Print):
                scan_expr(n.expr)
            elif isinstance(n, Return):
                scan_expr(n.expr)
```

```
256             elif isinstance(n, If):
257                 scan_expr(n.condition)
258                 scan_block(n.then_block)
259                 if n.else_block:
260                     scan_block(n.else_block)
261             elif isinstance(n, While):
262                 scan_expr(n.condition)
263                 scan_block(n.body)
264             elif isinstance(n, Block):
265                 scan_block(n)
266             elif isinstance(n, FunctionCall):
267                 for arg in n.args:
268                     scan_expr(arg)
269 
270         def scan_block(block: Block) -> None:
271             for stmt in block.statements:
272                 scan_stmt(stmt)
273 
274         scan_block(node.body)
275         return [(param, inferred.get(param, LanguageType.INTEGER)) for
               param in node.params]
276 
277     def _assign_variable_type(self, scope: SymbolTable, node: Assign,
           value_type: LanguageType) -> None:
278         try:
279             symbol = scope.lookup(node.name)
280         except SymbolTableError:
281             scope.define_variable(node.name, value_type, initialized=
                   True)
282             return
283 
284         if not isinstance(symbol, VariableSymbol):
285             raise TypeCheckError(self._error_at(node, f"'{node.name}'
                   is not a variable"))
286         if symbol.symbol_type != value_type:
287             raise TypeCheckError(
288                 self._error_at(
289                     node,
290                     f"Cannot assign {value_type.value} to variable '{
                           node.name}' of type {symbol.symbol_type.value}",
291                 )
292             )
293         symbol.initialized = True
294 
295     def _infer_binary_type(
296         self,
297         node: BinaryOp,
298         left_type: LanguageType,
299         right_type: LanguageType,
300     ) -> LanguageType:
301         # Integer operators: both operands must be Integer
302         if node.op in {"+", "-", "*"}:
303             if left_type != LanguageType.INTEGER or right_type !=
```

```
                LanguageType.INTEGER:
304                 raise TypeCheckError(
305                     self._error_at(node, f"Operator '{node.op}'
                        requires two Integer operands")
306                 )
307             return LanguageType.INTEGER
308 
309         # Integer division: both operands must be Integer, result is
            Integer
310         if node.op == "/":
311             if left_type != LanguageType.INTEGER or right_type !=
                LanguageType.INTEGER:
312                 raise TypeCheckError(
313                     self._error_at(node, f"Operator '{node.op}'
                        requires two Integer operands")
314                 )
315             return LanguageType.INTEGER
316 
317         # Float operators: both operands must be Float
318         if node.op in {"+.", "-.", "*.", "/."}:
319             if left_type != LanguageType.FLOAT or right_type !=
                LanguageType.FLOAT:
320                 raise TypeCheckError(
321                     self._error_at(node, f"Operator '{node.op}'
                        requires two Float operands")
322                 )
323             return LanguageType.FLOAT
324 
325         if node.op in {"==", "!="}:
326             if not self._is_numeric(left_type) or not self._is_numeric(
                right_type):
327                 raise TypeCheckError(
328                     self._error_at(
329                         node,
330                         f"Operator '{node.op}' requires two arithmetic
                            operands",
331                     )
332                 )
333             if left_type != right_type:
334                 raise TypeCheckError(
335                     self._error_at(
336                         node,
337                         f"Operator '{node.op}' requires operands of the
                             same type, got {left_type.value} and {
                            right_type.value}",
338                     )
339                 )
340             return LanguageType.BOOLEAN
341 
342         raise TypeCheckError(self._error_at(node, f"Unsupported
            operator: {node.op}"))
343 
344     def _lookup_function_symbol(self, node: FunctionCall) ->
```

```
        FunctionSymbol:
            try:
                symbol = self.global_scope.lookup(node.name)
            except SymbolTableError as error:
                raise TypeCheckError(self._error_at(node, str(error))) from
                    error
            if not isinstance(symbol, FunctionSymbol):
                raise TypeCheckError(self._error_at(node, f"'{node.name}'
                    is not a function"))
            return symbol

    @staticmethod
    def _is_numeric(language_type: LanguageType) -> bool:
        return language_type in {LanguageType.INTEGER, LanguageType.
            FLOAT}

    @staticmethod
    def _literal_type(node: Literal) -> LanguageType:
        if isinstance(node.value, bool):
            return LanguageType.BOOLEAN
        if isinstance(node.value, int):
            return LanguageType.INTEGER
        if isinstance(node.value, float):
            return LanguageType.FLOAT
        if isinstance(node.value, str):
            return LanguageType.STRING
        raise TypeCheckError(f"Unsupported literal value: {node.value!r
            }")

    @staticmethod
    def _error_at(node: ASTNode, message: str) -> str:
        return f"[line {node.line}, col {node.column}] TypeError: {
            message}"
```

Listing 8.8: components/type_checker.py — static type checker with inference

```
from __future__ import annotations

from dataclasses import dataclass
from typing import Callable

from components.ast_nodes import (
    ASTNode,
    Assign,
    BinaryOp,
    Block,
    FunctionCall,
    FunctionDef,
    Identifier,
    If,
    Literal,
    Print,
    Program,
```

```
    Return,
    While,
)
from components.symbol_table import LanguageType


class InterpreterError(Exception):
    pass


class _ReturnSignal(Exception):
    def __init__(self, value: object) -> None:
        self.value = value


@dataclass
class _FunctionRuntime:
    node: FunctionDef


class _Environment:
    def __init__(self, parent: _Environment | None = None) -> None:
        self.parent = parent
        self.values: dict[str, object] = {}

    def define(self, name: str, value: object) -> None:
        self.values[name] = value

    def get(self, name: str) -> object:
        if name in self.values:
            return self.values[name]
        if self.parent is not None:
            return self.parent.get(name)
        raise InterpreterError(f"Undefined variable: {name}")

    def assign(self, name: str, value: object) -> None:
        if name in self.values:
            self.values[name] = value
            return
        if self.parent is not None and self.parent.contains(name):
            self.parent.assign(name, value)
            return
        self.values[name] = value

    def contains(self, name: str) -> bool:
        if name in self.values:
            return True
        if self.parent is not None:
            return self.parent.contains(name)
        return False


class Interpreter:
```

```
    def __init__(self, output_callback: Callable[[str], None] | None =
        None) -> None:
        self._functions: dict[str, _FunctionRuntime] = {}
        self._output_callback = output_callback or print

    def execute(self, program: Program) -> None:
        environment = _Environment()
        for statement in program.statements:
            if isinstance(statement, FunctionDef):
                self._functions[statement.name] = _FunctionRuntime(node
                    =statement)

        for statement in program.statements:
            if isinstance(statement, FunctionDef):
                continue
            self._execute_statement(statement, environment)

    def _execute_statement(self, node: ASTNode, environment:
        _Environment) -> None:
        if isinstance(node, Assign):
            value = self._evaluate(node.value, environment)
            environment.assign(node.name, value)
            return

        if isinstance(node, Print):
            value = self._evaluate(node.expr, environment)
            self._output_callback(f"{self._format_value(value)} : {self
                ._runtime_type(value).value}")
            return

        if isinstance(node, Return):
            raise _ReturnSignal(self._evaluate(node.expr, environment))

        if isinstance(node, If):
            condition = self._evaluate(node.condition, environment)
            if not isinstance(condition, bool):
                raise InterpreterError(self._error_at(node.condition, "
                    If condition must evaluate to Boolean"))
            branch = node.then_block if condition else node.else_block
            if branch is not None:
                self._execute_block(branch, _Environment(parent=
                    environment))
            return

        if isinstance(node, While):
            while True:
                condition = self._evaluate(node.condition, environment)
                if not isinstance(condition, bool):
                    raise InterpreterError(self._error_at(node.
                        condition, "While condition must evaluate to
                        Boolean"))
                if not condition:
                    break
```

```
                self._execute_block(node.body, _Environment(parent=
                    environment))
            return

        if isinstance(node, Block):
            self._execute_block(node, _Environment(parent=environment))
            return

        if isinstance(node, FunctionCall):
            self._evaluate(node, environment)
            return

        raise InterpreterError(self._error_at(node, f"Unsupported
            statement node: {type(node).__name__}"))

    def _execute_block(self, block: Block, environment: _Environment)
        -> None:
        for statement in block.statements:
            self._execute_statement(statement, environment)

    def _evaluate(self, node: ASTNode, environment: _Environment) ->
        object:
        if isinstance(node, Literal):
            return node.value

        if isinstance(node, Identifier):
            return environment.get(node.name)

        if isinstance(node, BinaryOp):
            left_value = self._evaluate(node.left, environment)
            right_value = self._evaluate(node.right, environment)
            return self._evaluate_binary(node, left_value, right_value)

        if isinstance(node, FunctionCall):
            return self._call_function(node, environment)

        raise InterpreterError(self._error_at(node, f"Unsupported
            expression node: {type(node).__name__}"))

    def _call_function(self, node: FunctionCall, environment:
        _Environment) -> object:
        function_runtime = self._functions.get(node.name)
        if function_runtime is None:
            raise InterpreterError(self._error_at(node, f"Undefined
                function: {node.name}"))

        if len(node.args) != len(function_runtime.node.params):
            raise InterpreterError(
                self._error_at(
                    node,
                    f"Function '{node.name}' expects {len(
                        function_runtime.node.params)} argument(s), got
                        {len(node.args)}",
```

```
                )
            )

        call_environment = _Environment(parent=environment)
        argument_values = [self._evaluate(argument, environment) for argument in node.args]
        for parameter_name, argument_value in zip(function_runtime.node.params, argument_values, strict=False):
            call_environment.define(parameter_name, argument_value)

        try:
            self._execute_block(function_runtime.node.body, call_environment)
        except _ReturnSignal as signal:
            return signal.value
        return None

    def _evaluate_binary(self, node: BinaryOp, left_value: object, right_value: object) -> object:
        if node.op == "+":
            return left_value + right_value
        if node.op == "-":
            return left_value - right_value
        if node.op == "*":
            return left_value * right_value
        if node.op == "/":
            if int(right_value) == 0:
                raise InterpreterError(self._error_at(node, "Division by zero"))
            return int(left_value) // int(right_value)
        if node.op == "+.":
            return float(left_value) + float(right_value)
        if node.op == "-.":
            return float(left_value) - float(right_value)
        if node.op == "*.":
            return float(left_value) * float(right_value)
        if node.op == "/.":
            if float(right_value) == 0.0:
                raise InterpreterError(self._error_at(node, "Division by zero"))
            return float(left_value) / float(right_value)
        if node.op == "==":
            return left_value == right_value
        if node.op == "!=":
            return left_value != right_value
        raise InterpreterError(self._error_at(node, f"Unsupported operator: {node.op}"))

    @staticmethod
    def _format_value(value: object) -> str:
        if isinstance(value, str):
            return f'"{value}"'
        if isinstance(value, bool):
```

```
            return "true" if value else "false"
        return str(value)

    @staticmethod
    def _runtime_type(value: object) -> LanguageType:
        if isinstance(value, bool):
            return LanguageType.BOOLEAN
        if isinstance(value, int):
            return LanguageType.INTEGER
        if isinstance(value, float):
            return LanguageType.FLOAT
        if isinstance(value, str):
            return LanguageType.STRING
        return LanguageType.VOID

    @staticmethod
    def _error_at(node: ASTNode, message: str) -> str:
        return f"[line {node.line}, col {node.column}] RuntimeError: {
            message}"
```

Listing 8.9: components/interpreter.py — tree-walking interpreter

```
from __future__ import annotations

from dataclasses import dataclass

from components.ast_printer import ASTPrinter
from components.interpreter import Interpreter
from components.lexica import Lexer
from components.parser import Parser
from components.symbol_table import SymbolTable
from components.tokens import Token
from components.type_checker import TypeChecker


@dataclass
class PipelineResult:
    tokens: list[Token]
    ast_text: str
    checked_scope: SymbolTable
    outputs: list[str]


def run_pipeline(source_code: str) -> PipelineResult:
    tokens = Lexer(source_code).tokenize()
    tree = Parser(tokens).parse()
    ast_text = ASTPrinter().print(tree)
    checked_scope = TypeChecker().check(tree)

    outputs: list[str] = []
    Interpreter(output_callback=outputs.append).execute(tree)

    return PipelineResult(
```

```
        tokens=tokens,
        ast_text=ast_text,
        checked_scope=checked_scope,
        outputs=outputs,
    )


def format_tokens(tokens: list[Token]) -> str:
    lines = []
    for token in tokens:
        lines.append(
            f"  {token.token_type.name:<14} {token.lexeme!r:<12} line={token.line} col={token.column}"
        )
    return "\n".join(lines)


def format_stage_output(result: PipelineResult) -> str:
    sections = [
        "=" * 60,
        "  STAGE 1 -- TOKENS",
        "=" * 60,
        format_tokens(result.tokens),
        "",
        "=" * 60,
        "  STAGE 2 -- AST (parser output)",
        "=" * 60,
        result.ast_text,
        "",
        "=" * 60,
        "  STAGE 3 -- TYPE CHECK",
        "=" * 60,
        result.checked_scope.format_table(),
        "",
        "=" * 60,
        "  STAGE 4 -- EXECUTION",
        "=" * 60,
        "\n".join(result.outputs) if result.outputs else "(no output)",
    ]
    return "\n".join(sections)
```

Listing 8.10: components/pipeline.py — shared pipeline (CLI, UI, and tests)

A.3 Test Suite

```
from __future__ import annotations

import unittest

from components.lexica import Lexer, LexerError
from components.parser import Parser, ParseError
from components.pipeline import run_pipeline
```

```
from components.symbol_table import LanguageType, SymbolTable,
    SymbolTableError
from components.tokens import TokenType
from components.type_checker import TypeCheckError


#
    ------------------------------------------------------------------------

# Lexer
#
    ------------------------------------------------------------------------


class LexerTests(unittest.TestCase):
    """Tests for components/lexica.py -- lexical analysis and
       tokenisation."""

    def test_integer_literal_token(self) -> None:
        tokens = Lexer("42").tokenize()
        self.assertEqual(tokens[0].token_type, TokenType.INT_LITERAL)
        self.assertEqual(tokens[0].lexeme, "42")

    def test_float_literal_token(self) -> None:
        tokens = Lexer("3.14").tokenize()
        self.assertEqual(tokens[0].token_type, TokenType.FLOAT_LITERAL)
        self.assertEqual(tokens[0].lexeme, "3.14")

    def test_bool_literals_produce_bool_literal_tokens(self) -> None:
        tokens = Lexer("true false").tokenize()
        self.assertEqual(tokens[0].token_type, TokenType.BOOL_LITERAL)
        self.assertEqual(tokens[0].lexeme, "true")
        self.assertEqual(tokens[1].token_type, TokenType.BOOL_LITERAL)
        self.assertEqual(tokens[1].lexeme, "false")

    def test_keywords_produce_correct_token_types(self) -> None:
        tokens = Lexer("if else while def return print").tokenize()
        expected = [
            TokenType.IF, TokenType.ELSE, TokenType.WHILE,
            TokenType.DEF, TokenType.RETURN, TokenType.PRINT,
        ]
        actual = [t.token_type for t in tokens if t.token_type !=
           TokenType.EOF]
        self.assertEqual(actual, expected)

    def test_identifier_not_confused_with_keyword(self) -> None:
        tokens = Lexer("iffy whileloop").tokenize()
        self.assertEqual(tokens[0].token_type, TokenType.IDENTIFIER)
        self.assertEqual(tokens[1].token_type, TokenType.IDENTIFIER)

    def test_two_character_operators(self) -> None:
        tokens = Lexer("== !=").tokenize()
        self.assertEqual(tokens[0].token_type, TokenType.EQUAL_EQUAL)
```

```
        self.assertEqual(tokens[1].token_type, TokenType.BANG_EQUAL)

    def test_float_dot_operators_tokenised(self) -> None:
        tokens = Lexer("+. -. *. /.").tokenize()
        expected = [TokenType.PLUS_DOT, TokenType.MINUS_DOT, TokenType.
            STAR_DOT, TokenType.SLASH_DOT]
        actual = [t.token_type for t in tokens if t.token_type !=
            TokenType.EOF]
        self.assertEqual(actual, expected)

    def test_float_literal_not_confused_with_dot_operator(self) -> None
        :
        # "3.14" must produce FLOAT_LITERAL, not INT_LITERAL followed
            by something
        tokens = Lexer("3.14").tokenize()
        self.assertEqual(tokens[0].token_type, TokenType.FLOAT_LITERAL)
        self.assertEqual(tokens[0].lexeme, "3.14")

    def test_source_position_tracked_across_lines(self) -> None:
        tokens = Lexer("x\ny").tokenize()
        self.assertEqual(tokens[0].line, 1)
        self.assertEqual(tokens[1].line, 2)

    def test_unexpected_character_raises_lexer_error(self) -> None:
        with self.assertRaises(LexerError):
            Lexer("@").tokenize()

    def test_lone_exclamation_raises_lexer_error(self) -> None:
        with self.assertRaises(LexerError):
            Lexer("!5").tokenize()

    def test_unterminated_string_raises_lexer_error(self) -> None:
        with self.assertRaises(LexerError):
            Lexer('"hello').tokenize()


# ------------------------------------------------------------------------

# Symbol table
# ------------------------------------------------------------------------


class SymbolTableTests(unittest.TestCase):
    """Tests for components/symbol_table.py -- scoped symbol table and
        semantic types."""

    def test_define_variable_and_lookup(self) -> None:
        table = SymbolTable()
        table.define_variable("x", LanguageType.INTEGER, initialized=
            True)
        symbol = table.lookup("x")
```

```
97          self.assertEqual(symbol.symbol_type, LanguageType.INTEGER)
98
99      def test_duplicate_variable_definition_raises_error(self) -> None:
100         table = SymbolTable()
101         table.define_variable("x", LanguageType.INTEGER)
102         with self.assertRaises(SymbolTableError):
103             table.define_variable("x", LanguageType.FLOAT)
104
105     def test_duplicate_function_definition_raises_error(self) -> None:
106         table = SymbolTable()
107         table.define_function("f", LanguageType.INTEGER, [])
108         with self.assertRaises(SymbolTableError):
109             table.define_function("f", LanguageType.FLOAT, [])
110
111     def test_lookup_in_parent_scope(self) -> None:
112         parent = SymbolTable()
113         parent.define_variable("x", LanguageType.INTEGER, initialized=
                True)
114         child = parent.child_scope()
115         symbol = child.lookup("x")
116         self.assertEqual(symbol.symbol_type, LanguageType.INTEGER)
117
118     def test_undefined_symbol_raises_error(self) -> None:
119         table = SymbolTable()
120         with self.assertRaises(SymbolTableError):
121             table.lookup("z")
122
123     def test_format_table_contains_variable_row(self) -> None:
124         table = SymbolTable()
125         table.define_variable("counter", LanguageType.INTEGER,
                initialized=True)
126         output = table.format_table()
127         self.assertIn("counter", output)
128         self.assertIn("variable", output)
129         self.assertIn("Integer", output)
130
131     def test_format_table_contains_function_row(self) -> None:
132         table = SymbolTable()
133         table.define_function("add", LanguageType.INTEGER, [("a",
                LanguageType.INTEGER)])
134         output = table.format_table()
135         self.assertIn("add", output)
136         self.assertIn("function", output)
137
138
139 #
        ------------------------------------------------------------------------

140 # Parser
141 #
        ------------------------------------------------------------------------

142
```

```
class ParserTests(unittest.TestCase):
    """Tests for components/parser.py -- recursive-descent parsing."""

    def _parse(self, source: str):
        return Parser(Lexer(source).tokenize()).parse()

    def test_missing_semicolon_raises_parse_error(self) -> None:
        with self.assertRaises(ParseError):
            self._parse("x = 5")

    def test_unclosed_brace_raises_parse_error(self) -> None:
        with self.assertRaises(ParseError):
            self._parse("if (true) { x = 1;")

    def test_operator_precedence_mul_before_add(self) -> None:
        # 2 + 3 * 4 must be 14, not 20
        result = run_pipeline("x = 2 + 3 * 4; print(x);")
        self.assertEqual(result.outputs, ["14 : Integer"])

    def test_operator_left_associativity(self) -> None:
        # 10 - 3 - 2 must be 5 (left-assoc), not 9
        result = run_pipeline("x = 10 - 3 - 2; print(x);")
        self.assertEqual(result.outputs, ["5 : Integer"])

    def test_parentheses_override_precedence(self) -> None:
        # (2 + 3) * 4 must be 20
        result = run_pipeline("x = (2 + 3) * 4; print(x);")
        self.assertEqual(result.outputs, ["20 : Integer"])


# ---------------------------------------------------------------------------

# Type checker
# ---------------------------------------------------------------------------


class TypeCheckerTests(unittest.TestCase):
    """Tests for components/type_checker.py -- static type inference and
        checking."""

    def test_integer_arithmetic_stays_integer(self) -> None:
        result = run_pipeline("x = 3 + 4; print(x);")
        self.assertEqual(result.outputs, ["7 : Integer"])

    def test_division_always_produces_integer(self) -> None:
        # integer / integer -> Integer (no implicit float conversion)
        result = run_pipeline("x = 10 / 2; print(x);")
        self.assertEqual(result.outputs, ["5 : Integer"])

    def test_float_dot_operators_produce_float(self) -> None:
        result = run_pipeline("x = 1.0 +. 2.5; print(x);")
```

```
        self.assertEqual(result.outputs, ["3.5 : Float"])

    def test_mixed_int_float_addition_raises_type_error(self) -> None:
        # Integer + Float is no longer valid; must use +.
        with self.assertRaises(TypeCheckError):
            run_pipeline("x = 1 + 2.5; print(x);")

    def test_string_variable_type_inferred(self) -> None:
        result = run_pipeline('x = "hello"; print(x);')
        self.assertEqual(result.outputs, ['"hello" : String'])

    def test_boolean_literal_type_inferred(self) -> None:
        result = run_pipeline("x = true; print(x);")
        self.assertEqual(result.outputs, ["true : Boolean"])

    def test_boolean_equality_raises_type_error(self) -> None:
        # Boolean == Boolean is not allowed; == only accepts arithmetic
            operands
        with self.assertRaises(TypeCheckError):
            run_pipeline('x = true; if (x == false) { print("then"); }
               else { print("else"); }')

    def test_mixed_int_float_equality_raises_type_error(self) -> None:
        # == and != must not accept mixed Integer/Float (no overloading
           )
        with self.assertRaises(TypeCheckError):
            run_pipeline("x = 5; y = 3.14; if (x == y) { print(x); }")

    def test_mixed_int_float_inequality_raises_type_error(self) -> None
       :
        with self.assertRaises(TypeCheckError):
            run_pipeline("x = 5; y = 3.14; if (x != y) { print(x); }")

    def test_arithmetic_on_string_raises_type_error(self) -> None:
        with self.assertRaises(TypeCheckError):
            run_pipeline('x = "a" + 1;')

    def test_if_non_boolean_condition_raises_type_error(self) -> None:
        with self.assertRaises(TypeCheckError):
            run_pipeline("if (1) { x = 2; }")

    def test_while_non_boolean_condition_raises_type_error(self) ->
       None:
        with self.assertRaises(TypeCheckError):
            run_pipeline("x = 0; while (x) { x = x + 1; }")

    def test_assignment_type_mismatch_raises_type_error(self) -> None:
        with self.assertRaises(TypeCheckError):
            run_pipeline('x = 5; x = "hello";')

    def test_undefined_variable_raises_type_error(self) -> None:
        with self.assertRaises(TypeCheckError):
            run_pipeline("print(z);")
```

```
239
240     def test_function_wrong_arg_count_raises_type_error(self) -> None:
241         with self.assertRaises(TypeCheckError):
242             run_pipeline("def f(a) { return a; } f(1, 2);")
243
244     def test_function_arg_type_mismatch_raises_type_error(self) -> None
        :
245         # first call infers a: Integer; second call with String must
            fail
246         with self.assertRaises(TypeCheckError):
247             run_pipeline('def f(a) { return a; } f(1); f("hello");')
248
249     def test_uncalled_function_body_type_error_is_caught(self) -> None:
250         # type error inside body must be detected even if function is
            never called
251         with self.assertRaises(TypeCheckError):
252             run_pipeline('def bad() { y = "hello" + 1; }')
253
254     def test_valid_uncalled_function_does_not_raise(self) -> None:
255         # a well-typed but uncalled function should not raise
256         result = run_pipeline("def add(a, b) { return a + b; }")
257         self.assertEqual(result.outputs, [])
258
259
260 #
    ------------------------------------------------------------------------

261 # Integration (full pipeline)
262 #
    ------------------------------------------------------------------------

263
264 class IntegrationTests(unittest.TestCase):
265     """End-to-end tests through all four pipeline stages."""
266
267     # --- Unary minus ---
268
269     def test_unary_minus_integer(self) -> None:
270         result = run_pipeline("x = -5; print(x);")
271         self.assertEqual(result.outputs, ["-5 : Integer"])
272
273     def test_unary_minus_float(self) -> None:
274         result = run_pipeline("y = -. 3.14; print(y);")
275         self.assertEqual(result.outputs, ["-3.14 : Float"])
276
277     def test_unary_minus_variable(self) -> None:
278         result = run_pipeline("x = 10; y = -x; print(y);")
279         self.assertEqual(result.outputs, ["-10 : Integer"])
280
281     def test_unary_minus_inside_expression(self) -> None:
282         result = run_pipeline("x = 3 + -2; print(x);")
283         self.assertEqual(result.outputs, ["1 : Integer"])
284
```

```
    # --- Control flow ---

    def test_if_then_branch_executes(self) -> None:
        result = run_pipeline(
            'x = 5; if (x == 5) { print("yes"); } else { print("no"); }'
        )
        self.assertEqual(result.outputs, ['"yes" : String'])

    def test_if_else_branch_executes_when_condition_false(self) -> None:
        result = run_pipeline(
            'x = 0; if (x == 5) { print("yes"); } else { print("no"); }'
        )
        self.assertEqual(result.outputs, ['"no" : String'])

    def test_while_loop_executes_repeatedly(self) -> None:
        result = run_pipeline("x = 3; while (x != 0) { print(x); x = x - 1; }")
        self.assertEqual(result.outputs, ["3 : Integer", "2 : Integer", "1 : Integer"])

    # --- Functions ---

    def test_function_integer_return(self) -> None:
        result = run_pipeline("def square(n) { return n * n; } print(square(7));")
        self.assertEqual(result.outputs, ["49 : Integer"])

    def test_function_type_inference(self) -> None:
        result = run_pipeline(
            "def add(a, b) { return a +. b; } result = add(2.0, 3.5); print(result);"
        )
        self.assertEqual(result.outputs, ["5.5 : Float"])
        self.assertIn("result       variable   Float", result.checked_scope.format_table())

    def test_function_value_parameter_passing(self) -> None:
        # Modifying a parameter inside a function must not affect the caller's variable
        result = run_pipeline(
            "def inc(a) { a = a + 1; return a; } x = 10; y = inc(x); print(x); print(y);"
        )
        self.assertEqual(result.outputs, ["10 : Integer", "11 : Integer"])

    def test_nested_function_calls(self) -> None:
        result = run_pipeline(
            "def add(a, b) { return a + b; } print(add(add(1, 2), 3));"
        )
```

```
        self.assertEqual(result.outputs, ["6 : Integer"])

    # --- print() with all four types ---

    def test_print_integer(self) -> None:
        result = run_pipeline("print(42);")
        self.assertEqual(result.outputs, ["42 : Integer"])

    def test_print_float(self) -> None:
        result = run_pipeline("print(3.14);")
        self.assertEqual(result.outputs, ["3.14 : Float"])

    def test_print_boolean(self) -> None:
        result = run_pipeline("print(true);")
        self.assertEqual(result.outputs, ["true : Boolean"])

    def test_print_string(self) -> None:
        result = run_pipeline('print("plc");')
        self.assertEqual(result.outputs, ['"plc" : String'])

    # --- Reassignment ---

    def test_variable_can_be_reassigned_same_type(self) -> None:
        result = run_pipeline("x = 1; x = 2; x = 3; print(x);")
        self.assertEqual(result.outputs, ["3 : Integer"])

    def test_scope_isolation_if_block(self) -> None:
        # Variable defined inside an if-block must not be visible
            outside it
        with self.assertRaises(TypeCheckError):
            run_pipeline(
                "x = 1; if (x == 1) { inner = 99; } print(inner);"
            )


if __name__ == "__main__":
    unittest.main()
```

Listing 8.11: tests/test_pipeline.py — automated regression suite (59 tests)